University of Amsterdam

Master's Thesis

Interventions in Asymmetric Belief Systems

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A thesis submitted in partial fulfilment of the requirements for the degree of Master of Science in Computational Science

in the

Computational Science Lab
Informatics Institute

July 2026

Declaration of Authorship

I, Henry Zwart, declare that this thesis, entitled 'Interventions in Asymmetric Belief Systems' and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at the University of Amsterdam.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
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Abstract

Faculty of Science
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Master of Science in Computational Science

Interventions in Asymmetric Belief Systems

by Henry ZWART

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Acknowledgements

Thank the people that have helped: supervisors, family, etc.

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Abbreviations

CSL **C**omputational **S**cience **L**ab

UvA **U**niversiteit **v**an **A**msterdam

Chapter 1

Terminology and notation

Define key terminology, outline notation used in following sections.

Chapter 2

Introduction

TODO:

- Address the specific variables we look at in the outbound/inbound experiments. Are the variables interesting independently of the asymmetry results/theoretically motivated?
- Mention our hypotheses, where they exist.
  - RQ3: That interventions can only be effective when the target and the point-of-intervention are both low.

2.1 Plan

- Motivate the problem
- Contributions
- Research questions (thinking these are perhaps better left to later, in favour of contributions here)

The probable existence of directed causal relations between beliefs and attitudes, as well as their potential implications for endogenous dynamics, is widely acknowledged in recent studies on belief system modelling. In spite of this, empirical evidence for the existence and impacts of such directed relations remains scarce, likely due, in-part, to a historical focus on bidirectional (i.e., symmetric) models in theories of belief dynamics (**CITE**), as well as the additional data requirements associated with inferring directional relations (Nguyen et al., 2017, p. 35) (**CITE**).

These unknowns, *existence* and *impact*, are the primary subjects of this thesis, and of our first two research questions, introduced back in Chapter 2:

RQ1: To what extent are causal relations *symmetric* or *asymmetric*, in models of climate change belief systems inferred from the climate beliefs dataset?

RQ2: How do asymmetric and symmetric beliefs systems differ with regards to intervention strategy and effectiveness, in models inferred from the climate beliefs dataset?

The *climate beliefs dataset*, here, refers to the reduced longitudinal dataset whose construction we describe in Chapter 9.3. These research questions are concerned with population-level belief systems and intervention effects. However, while some aspects of belief system structure are likely shared within a population, the notion of a belief system is inherently individual. We also expect differences in individuals' responses to interventions. Our final research questions address these two topics, specifically for asymmetric belief systems:

RQ3: How do intervention outcome and effectiveness vary between individuals with different initial conditions in asymmetric belief systems inferred from the climate beliefs dataset?

RQ4: How do asymmetric belief systems inferred from the climate beliefs dataset vary between conservative and liberal individuals?

2.2 Motivation

Issue support on climate policies in US driven by political identification and climate beliefs (Bumann, 2021; Roser-Renouf et al., 2014; Shao & Hao, 2020; Unsworth & Fielding, 2014; Ziegler, 2017).

But political identification is often not consistent with policy attitudes (Huddy et al., 2015; Iyengar et al., 2012). More often identity-driven (symbolic) than issue-driven (operational) (Mason, 2018). Also (Egan, 2020).

Suggestions of asymmetric/causal directionality in belief systems:
- So far reviewed in Zotero: Belief Networks, Papers from Sara
- (Brandt et al., 2019, p. 2,9,10)
- (Brandt, 2022, p. 3,22)
- (Keskintürk, 2022, p. 10)
- (van Noord et al., 2025, p. 4)
- (Brandt & Sleegers, 2021, p. 2): Belief system *dynamics* require causal connections.
  - Constraint, causality, exogenous factors all necessary for any theory of BS dynamics.
  - Though still interprets edges as undirected/bi-directional
- (Brandt & Sleegers, 2021, p. 22): "In some cases … assume that causal influence for some elements (e.g., partisan identification) is primarily in one direction".
- (Converse, 2006, p. 208) (mentioned in (Brandt & Sleegers, 2021, p. 2))
- (Coppock & Green, 2022): Referenced in (Brandt & Sleegers, 2021)

Self-interaction effects:
- (Brandt & Sleegers, 2021, p. 22): "we made the simplifying assumption that a node does not affect itself".

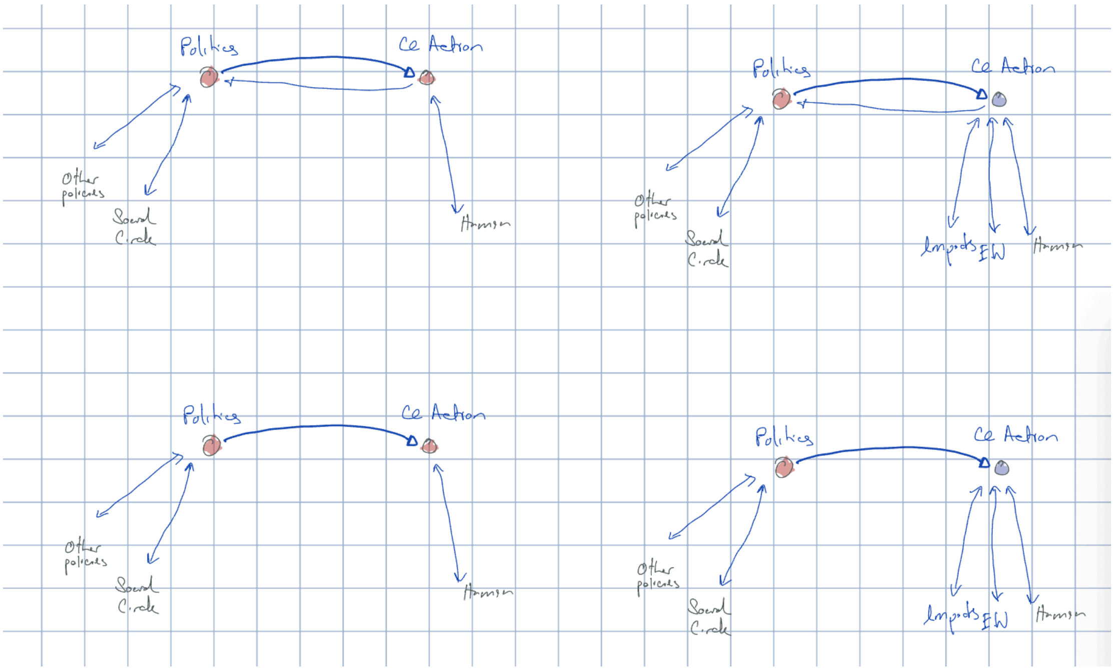
Variation in belief systems between individuals:
- (Brandt & Sleegers, 2021, p. 4,20) "connections will likely vary between people, time, and political contexts", "Although … belief systems are at the individuals level, this does not mean that structure is not shared"

Stability of belief dynamics:
- (Osborne & Sibley, 2020)
- (Kiley & Vaisey, 2020)

2.3 Asymmetry example

Suppose thin arrows have weight 1, thick arrows have weight 2, and a belief/attitude adopts the dominant state in its neighbourhood, weighted by the incoming edge weights. If your support for non-climate-related policies and your social circle are Republican-aligned, you adopt that political ideology. Suppose that you believe in human-caused climate change, then that reinforces your support for climate action; however, the net support is -1, so you flip, taking the support for Republican politics to $+4$. If you then become concerned about extreme weather, and believe that the impacts of climate change are high, your support for climate action flips to positive with a net support of +1. However, this is not sufficient to shift your Republican alignment, which stays at −1. In the extreme case, where there is no feedback to Politics, your change in attitude has no bearing on your political alignment.

Asymmetry is then best thought of as the weight imposed on one belief/attitude by another being different from the opposite direction. To overcome Republican alignment, we need at least two pro-Democrat attitudes. To overcome negative `CC Action`, we require at least three other pro-climate attitudes.



2.4 Research questions

Theoretical contributions:

- **RF1.1:** Extending the causal attitude network model of belief systems to: (i) support asymmetric causal effects between beliefs and attitudes, and (ii) model intervention dynamics.

Inferring belief systems:

- **RQ2.1:** To what extent are causal relations *symmetric* or *asymmetric* in models of climate change belief systems inferred from observational data?

- **RQ2.2:** How do (symmetric or asymmetric) belief systems relating to climate change vary between conservative and liberal individuals?

Intervention dynamics:

- **RQ3.1:** How do asymmetric and symmetric beliefs systems inferred from the climate attitudes dataset differ with regards to intervention strategy and effectiveness?

- **RQ3.2:** How do intervention outcome and effectiveness vary between individuals with different initial conditions, or between conservative and liberal individuals?

2.5 Contributions

1. We present a mathematical model for belief system dynamics that does not assume equilibrium and does not assume symmetric influence between cognitive aspects (Chapter 3)
2. We describe a novel parameter estimation method for fitting binary Ising models to continuous data (Chapter 4).
3. We calibrate said model to data from a recent longitudinal survey including items on beliefs and attitudes regarding climate change (Chapter 6).
4. We demonstrate, by way of the calibrated model, the existence of asymmetric influence relations between beliefs and attitudes. Furthermore we show that influence relations are not *necessarily* asymmetric, may vary in the degree of asymmetry, and can be unidirectional (Chapter 6).
5. We demonstrate that the decision to represent asymmetric relations in belief system models can change intervention dynamics, and therefore conclusions one draws regarding intervention effect and effectiveness (Chapter 6).
6. We then show that belief systems may vary significantly between individuals, by fitting the proposed model to subsets of the climate attitudes dataset comprising conservative and liberal individuals (Chapter 7).
7. Finally, we show that reasoning about the effects of interventions on *individuals* is, in general, non-trivial. How an individual responds to an intervention typically depends on their prior belief system state, including beliefs and attitudes other than the target and goal of intervention (Chapter 7).

Chapter 3

Non-equilibrium belief systems

3.1 Belief system dynamics

We will now formalise the dynamics of an individual belief system, under the following assumptions:

1. Beliefs and attitudes can be treated as discrete entities.

2. The state of a belief or attitude is Markovian with respect to the previous system configurations.

3. The probability distribution relating current and future belief states does not change over time.

The first assumption includes two key points. Firstly, that it makes sense to consider beliefs and attitudes as *entities*, or *objects*. This contrasts, for instance, alternative interpretations of attitudes as emissions from an underlying structure. (**TODO:** review CAN paper for further discussion on this matter). Secondly, that beliefs and attitudes can be considered *discrete* or *separable*. This is to say that there are no absolute physical constraints on the combinations of states which may be observed (although certain combinations may be highly unlikely). In particular, this captures the requirement that each belief or attitude can be represented using a single state variable.

Our second assumption concerns the treatment of beliefs and attitudes as Markovian.

- Harder to justify conceptually, since individuals have memory and can recall prior states beyond the previous second.

Consider a belief or attitude S_i with domain Ω_{S_i}. The state of S_i at time t depends on the previous states of other beliefs and attitudes, and we can therefore describe the trajectory of S_i as a Markov process:

$$\{s_i^t\}_{t=1}^{\infty} \quad \text{where} \quad s_i^t \in \Omega_{S_i} \tag{3.1}$$

with conditional state probability $P\big(S_i^{\tau+1} \mid \boldsymbol{S}^1, ..., \boldsymbol{S}^\tau\big) = P\big(S_i^{\tau+1} \mid \boldsymbol{S}^\tau\big)$ for $\tau \in \mathbb{N}$.

Since the state of S_i^{t+1} depends only on the previous belief system state, it follows that for any other belief or attitude S_j, the states S_i^{t+1} and S_j^{t+1} are conditionally independent given $\boldsymbol{S}^t$. We can therefore straightforwardly extend Equation 3.1 to describe the evolution of the entire belief system as a Markov process,

$$\{\boldsymbol{s}^t\}_{t=1}^{\infty} \quad \text{where } \boldsymbol{s}^t \in \Omega_{S_1} \times \cdots \times \Omega_{S_n} \tag{3.2}$$

with the probability of each belief system configuration given by the conditional distribution $P\big(\boldsymbol{S}^{t+1} \mid \boldsymbol{S}^t\big)$.

Under this conceptualisation, the task of modelling a belief system thus reduces to describing how the instantaneous configuration of belief states affects the probability distribution over possible future states.

3.2 Non-equilibrium belief system model

When studying the *symmetric* Ising model it is common to assume that the model is at equilibrium, such that $P(\boldsymbol{S}^{t+1} = \boldsymbol{s} \mid \boldsymbol{S}^t) = P(\boldsymbol{S} = \boldsymbol{s})$ for all configurations $\boldsymbol{s}$ and observation times t. The probability of observing $\boldsymbol{s}$ is then time-invariant and given by the Boltzmann probability parameterised by the Hamiltonian $H(\boldsymbol{s})$ (Cardy, 1996; Christensen & Moloney, 2005):

$$P(\boldsymbol{S} = \boldsymbol{s}) = \frac{e^{-\frac{1}{k_b T} \cdot H(\boldsymbol{s})}}{\sum_{\boldsymbol{s}'} e^{-\frac{1}{k_b T} \cdot H(\boldsymbol{s}')}} \tag{3.3}$$

where k_b is the Boltzmann constant, and the temperature parameter $T \in \mathbb{R}^+$ controls stochasticity. Equation 3.3 converges to a uniform distribution over all configurations when $T \to +\infty$, and a uniform distribution over the restricted set of configurations for which $H(\boldsymbol{s}')$ is minimised when $T \to 0^+$.

For the purposes of this study, we are interested in intervention dynamics, which are inherently non-equilibrium. To see why this is the case, notice that the purpose of intervention is to change the distribution of observed states. This is true both interventions intended to change the dominant state (e.g., to promote a sustainable alternative to a typical unsustainable behaviour) or reinforce it. Any belief system model intended for studying intervention dynamics therefore cannot assume equilibrium, and cannot define model dynamics using the Boltzmann distribution.

We instead define the conditional distribution $P(\boldsymbol{S}^{t+1} = \boldsymbol{s} \mid \boldsymbol{S}^t)$ explicitly. Recall that the states of each pair of spins S_i, S_j at time t are conditionally independent random variables, given knowledge of the previous configuration $\boldsymbol{S}^t$ (Chapter 3.1). It therefore suffices for us to describe the distribution over spin states for an individual spin S_i at time $t+1$, as conditional on $\boldsymbol{S}^t$.

As in the standard Ising model formulation, we will assume that the model typically evolves in the direction of lower energy; however, rather than defining the energy at the level of the model using the Hamilonian, we consider the energy experienced by S_i as a result of its baseline activation (also known as a *local field effect*), and interactions with other spins.

The baseline activation $h_i \in \mathbb{R}$ describes the tendency for S_i to take on the value $+1$ in the absence of interactions with other spins, or, more generally, when pairwise spin interactions have a net-effect of zero. S_i tends to adopt the value $+1$ under these circumstances, iff, $h_i > 0$, and -1 iff $h_i < 0$. This tendency increases with the magnitude of h_i.

Other spins influence S_i's state through alignment or opposition relations. For a spin S_j, we define an interaction effect J_{ji} (read *'influence of j on i'*). When J_{ij} if positive or negative, S_i is influenced to adopt the state S_j^t or $-S_j^t$ respectively. In the special case when $j = i$, we refer to J_{ii} as a *self-influence* effect. Positive self-influence effects reflect the inertia of S_i, i.e., the tendency to sustain a particular belief or attitude irrespective of other beliefs and attitudes. Negative self-influence effects are not clearly interpretable. The inclusion of self-influence effects allows us to capture the timescales of different beliefs or attitudes independently.

The energy experienced by S_i for a particular spin state $s \in \{-1, +1\}$ is then the result of s in combination with the baseline activation and influence effects from all spins with edges to S_i:

$$H_i(s \mid \boldsymbol{s}^t) = h_i \cdot s + \sum_j A_{ji} J_{ji} s_j^t \cdot s \tag{3.4}$$

where $\boldsymbol{A} \in \{0,1\}^{n \times n}$ is a pairwise adjacency matrix, with $A_{ji} = 1$, iff, S_j influences S_i. Dividing the common factor of s, we can equivalently write Equation 3.4 as

$$H_i(s \mid \boldsymbol{s}^t) = s \cdot h_i^{\text{eff}}(\boldsymbol{s}^t) \tag{3.5}$$

where $h_i^{\text{eff}}(\boldsymbol{s}^t) = h_i + \sum_j A_{ji} J_{ji} s_j^t$ is the effective baseline activation experienced by S_i at time t. We then define the probability that S_i adopts the state s at time t as:

$$\begin{aligned} P\big(S_i^{t+1} = s \mid \boldsymbol{S}^t = \boldsymbol{s}\big) &= \frac{e^{\frac{1}{T} H_i(s \mid \boldsymbol{s})}}{\sum_{s' \in \{0,1\}} e^{\frac{1}{T} H_i(s' \mid \boldsymbol{s})}} \\ &= \frac{e^{\frac{1}{T} H_i(s \mid \boldsymbol{s})}}{e^{\frac{1}{T} H_i(s \mid \boldsymbol{s})} + e^{-\frac{1}{T} H_i(-s \mid \boldsymbol{s})}} \\ &= \frac{e^{\frac{1}{T} s \cdot h_i^{\text{eff}}(\boldsymbol{s})}}{e^{\frac{1}{T} s \cdot h_i^{\text{eff}}(\boldsymbol{s})} + e^{-\frac{1}{T} s \cdot h_i^{\text{eff}}(\boldsymbol{s})}} \end{aligned} \tag{3.6}$$

For $s = +1$, this reduces to the logistic function:

$$\begin{aligned} p_i^{\boldsymbol{s}} := P\big(S_i^{t+1} = +1 \mid \boldsymbol{S}^t = \boldsymbol{s}\big) &= \frac{e^{\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})}}{e^{\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})} + e^{-\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})}} \\ &= \frac{1}{1 + e^{-2 \cdot \frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})}} \\ &= \text{logistic}\big(2 h_i^{\text{eff}}(\boldsymbol{s})/T\big) \end{aligned} \tag{3.7}$$

Therefore the complete conditional distribution is given by:

$$\begin{aligned} P\big(\boldsymbol{S}^{t+1} = \boldsymbol{s}^{t+1} \mid \boldsymbol{S}^t = \boldsymbol{s}^t\big) &= \prod_{i=1}^{n} P\big(S_i^{t+1} = s_i \mid \boldsymbol{S}^t = \boldsymbol{s}^t\big) \\ &= \prod_{i=1}^{n} \left[\left(\frac{1 + s_i}{2}\right) p_i^{\boldsymbol{s}^t} + \left(\frac{1 - s_i}{2}\right)\left(1 - p_i^{\boldsymbol{s}^t}\right)\right] \end{aligned} \tag{3.8}$$

Where the initial distribution $P(\boldsymbol{S}^0)$ is specified explicitly.

3.3 Symmetric and asymmetric belief systems

Let $\mathcal{M}$ be a belief system model defined as in the preceding section, comprising $N \in \mathbb{N}$ beliefs or attitudes, with adjacency matrix $\boldsymbol{A} \in \{0,1\}^{N \times N}$, interaction effect matrix $\boldsymbol{J} \in \mathbb{R}^{N \times N}$, and baseline activations $\boldsymbol{h} \in \mathbb{R}^N$.

When each entry in $\boldsymbol{J}$ is independent of all others, we say that $\mathcal{M}$ is an **asymmetric belief system**. The term *asymmetric* here refers to the directed relations between a pair of nodes. In an asymmetric belief system, for any pair of distinct nodes $S_i \neq S_j$, it may be the case that an influence relation exists only in one direction, or that the directed relations differ in magnitude.

In the special case where we constrain $A_{ij} = A_{ji}$ and $J_{ij} = J_{ji}$ for all $i, j \in [1, N]$, we instead say that $\mathcal{M}$ is a **symmetric belief system**. In a symmetric belief system all interactions are directionally-equivalent. This can be considered a simple extension of the symmetric Ising model with (i) self-interaction terms and (ii) temporal dynamics, thus the symmetric belief system model tends toward an equilibrium steady state.

3.4 Simulation via Glauber dynamics

It is straightforward to draw samples from a belief system model using Glauber dynamics, given the conditional probability distribution defined in Equation 3.8.

Given an initial state $\boldsymbol{s}^0 \in \{-1, +1\}^N$, we sample a sequence of $T \in \mathbb{N}$ subsequent configurations:

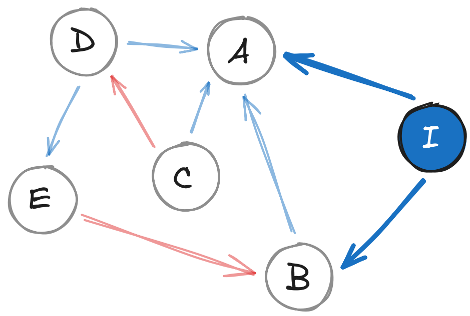
$$\{\boldsymbol{s}^t\}_{t=0}^{T}, \quad \text{where each } \boldsymbol{s}^{t+1} \sim P(\boldsymbol{S}^{t+1} \mid \boldsymbol{S}^t) \tag{3.9}$$

Each belief or attitude has the opportunity to update during every time interval. This update routine is referred to as *synchronous* Glauber dynamics, contrasting *asynchronous* Glauber dynamics, in which only one spin can update during a given interval.

3.5 Modelling interventions

We now outline our approach to modelling interventions in the asymmetric belief system model. In this study we consider interventions which affect the *state* of a belief system. Notably, this excludes interventions intended to influence the structure of a belief system, for instance, by changing the existence of effect size of influence relations between beliefs and attitudes.

Let $\mathcal{M}$ be a belief system model with parameters $\langle \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h} \rangle$. We can consider an intervention as an auxiliary node, I, in the belief system network, with state fixed at a particular value and outgoing edges toward a subset of beliefs and attitudes:



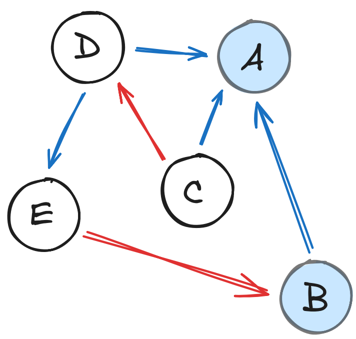
The intervention node I exerts influence on the belief system nodes A and B. Let us consider the effective baseline activation at node A and time $t+1$, as defined in Equation 3.4, given a previous configuration $\boldsymbol{s}^t$:

$$h_A^{\text{eff}}(\boldsymbol{s}^t) = h_A + \sum_{\text{node } j} A_{jA} J_{jA} s_j^t \tag{3.10}$$

Since the state of I is fixed (in this case, at $+1$), the contribution of the intervention node to A's effective activation baseline is constant. We can then re-write Equation 3.10, interpreting the intervention effect as an adjustment to the baseline activation h_A:

$$h_A^{\text{eff}}(\boldsymbol{s}^t) = (h_A + J_{IA}) + \sum_{\text{node } j \neq I} A_{jA} J_{jA} s_j^t \tag{3.11}$$

It follows then that we can model interventions more simply as an adjustment to the baseline activation of certain beliefs and attitudes:



Let us define this more formally. For a belief system model $\mathcal{M}$ with $N \in \mathbb{N}$ nodes, and parameters $\langle \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h} \rangle$, an *intervention*, is the function $\varphi_{\mathcal{M}} : \delta_h \mapsto \mathcal{M}'$, where $\delta_h \in \mathbb{R}^N$ is a vector of offsets to the baseline activations, and the model $\mathcal{M}'$ has parameters $\langle \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h}' \rangle$, where

$$\boldsymbol{h}' = \boldsymbol{h} + \delta_h \tag{3.12}$$

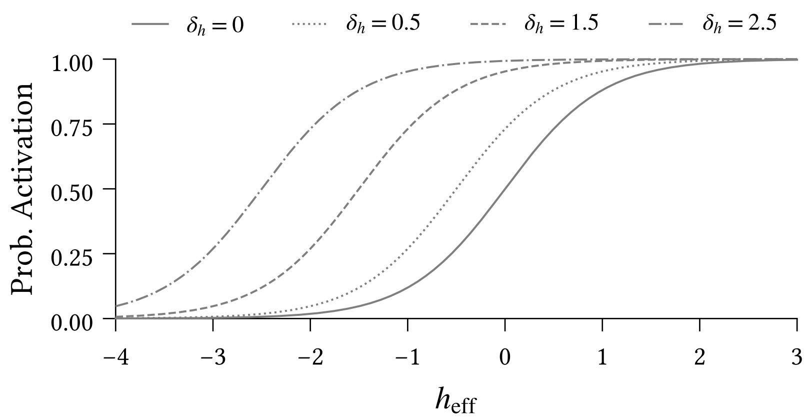


Figure 3.3: Impact of different intervention strengths ($\delta_h \in \{0, 0.5, 1.5, 2.5\}$) on activation probability for the intervention spin. Activation probability is calculated using Equation 3.7.

Figure 3.3 illustrates how the probability of activation for the intervention spin (Equation 3.7) changes for different intervention scenarios. Recall that a spin's activation probability is a function of it's effective local field. Since interventions constitute offsets to the effective local field, they are reflected as horizontal shifts in the logistic curve.

The solid line denotes the null scenario in which no intervention is applied. Consider two individuals, positioned at different locations on this base curve:

1. $h_{\text{eff}} = 0$: Ambivalent disposition toward the intervention spin, and

2. $h_{\text{eff}} = -2$: Strong negative disposition toward the intervention spin.

To understand the impacts of intervention strength on each individual, we draw a vertical line upward from the solid curve at each location, and read off the activation probability at each intersection with the other intervention curves. Under the null scenario the first individual exhibits an equal tendency toward each state, while the second is almost certain to adopt the state -1.

Notice that interventions are not experienced equivalently by the two individuals. For a weak intervention ($\delta_h = 0.5$), the first individual sees a substantial jump in activation probability, while the corresponding increase for the second individual is negligible. The stakes are reversed under a strong intervention ($\delta_h = 2.5$).

We can understand this difference in effect both conceptually and analytically. From a conceptual perspective, we can explain the large shift of the first individual under a weak intervention as resulting from their prior indifference, making them maximally susceptible to intervention. Prior to intervention, the second individual has a strong negative stance on this belief/attitude; weak interventions are insufficient to change their view. On the other hand, the second individual's negative stance provides more space for their attitude or belief to shift under a successful intervention. Compare this to the first individual, who is shifting from a place of indifference.

TODO: Quantitative explanation:

- Take derivative of the difference in activation probabilities between intervention and null scenario; solve for derivative equal to zero. Corresponds to the unique maximum since for $h_{\text{eff}} \to \pm\infty$ the second derivative is strictly positive.
- We find that maximum is at $h_{\text{eff}} = -\frac{\delta_h}{2}$.
- i.e., for a given intervention strength, the maximum change in activation probability occurs when the intervention causes h_{eff} to increase to same magnitude on the other side of zero.
- Individuals below this point require a stronger intervention to move the same amount.
- Individuals above this point experience ceiling effect. A smaller intervention could achieve almost the same effect.

Chapter 4

Methods

4.1 Counterfactual interventions

Individuals have different belief systems. Influence relations between beliefs and attitudes reflect heterogeneity in individual experiences and perception. Often these relations can themselves be considered beliefs — whether or not support for climate change mitigation leads to support for any specific policy depends in-part on an individual's belief regarding the policy's relevance. An individual's baseline activations, in turn, reflect a culmination of factors which determine their tendency toward certain beliefs or attitudes.

Our approach to parameter estimation, however, fundamentally assumes that the inferred belief system is shared by all individuals in the dataset from which the model is estimated. We note that this is not strictly a limitation of the parameter estimation method, nor of the model itself. Rather this assumption is imposed by data limitations. Fitting individual models requires substantial data from each individual, which is not available in the climate attitudes dataset. Moreover, even with substantial observational data, we are unlikely to observe all state transitions necessary to understand the *potential* dynamics of an individual system, owing to the relatively slow-moving nature of beliefs and attitudes.

For any given individual, the inferred models may thus not accurately reflect the stability of belief system states. A configuration which is stable in a given individual's own belief system may be considered unstable in the shared model. When simulating interventions, we take the most recent measurement for each individual as their initial state. However, due to this perceived instability, we expect to observe natural dynamics away from this initial state, even in null-intervention scenarios.

We thus consider intervention effects as relative to the null-intervention counterfactual baseline rather than relative to the initial state. For each intervention we run both intervention and null ($\delta_h = 0$) scenarios with identical random number generation settings, and define the *intervention effect* as the difference between the observed outcomes.

4.2 Binarisation

Prior to model fitting, we binarise all variables, mapping data values to the domain $\{-1, +1\}$. Before binarising, we shift each variable such that the *survey midpoint* (e.g., '3' on a 5-point Likert scale) aligns with 0, and rescale each variable such that the minimum and maximum possible values map to -1 and $+1$ respectively. For variables with no well-defined midpoint, such as those with an even number of possible responses, we shift such that the minimal and maximal values are at equal distance from 0.

We use a smooth binarisation process to robustly handle data points which are zero, or close to zero. This involves perturbing each data value $x \in \mathbb{R}$ by an independent noise term $\varepsilon \sim \mathcal{N}(0, \sigma)$ before thresholding. Figure 4.1 illustrates this process for a negative value of x and given choice of σ.

Observe that x is mapped to $+1$ if, and only if, ε is sufficiently large, such that $x + \varepsilon > 0$ (region A), or equivalently when $\varepsilon < x$ (region B). The probability that x is mapped to $+1$ is then

$$P(x \mapsto +1) = \Phi\left(\frac{x}{\sigma}\right) \tag{4.1}$$

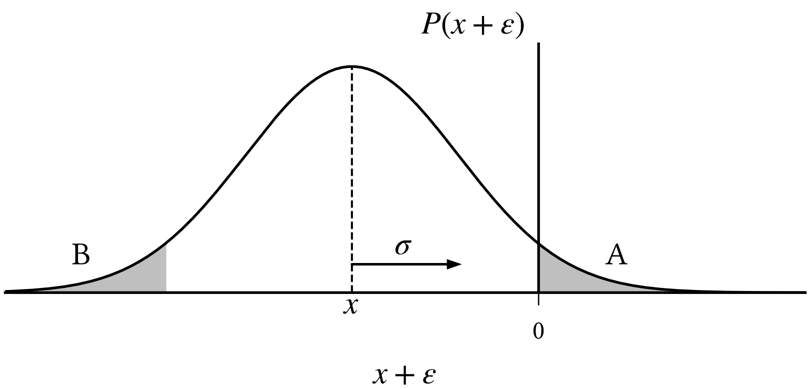


Figure 4.1: $x \in \mathbb{R}$ is smoothly binarised to $\{-1, +1\}$ by thresholding $x' = (x + \varepsilon)$ where $(x + \varepsilon) \sim \mathcal{N}(x, \sigma)$. Negative values are mapped to $+1$ with probability $P(x' > 0) = \text{A} = \text{B} = P(\varepsilon < x)$, which increases with σ and $|x|^{-1}$.

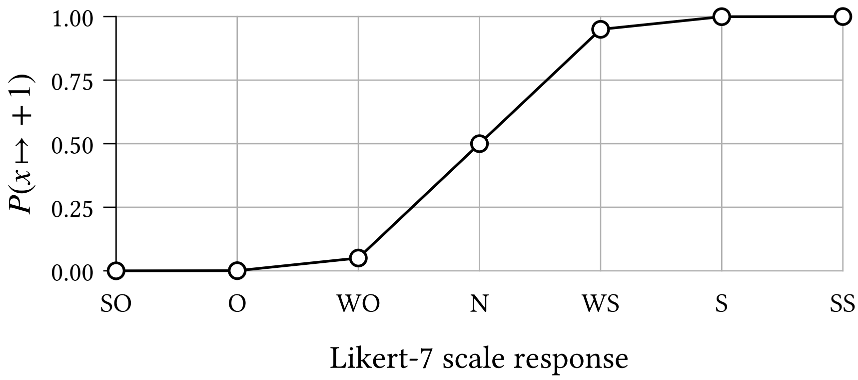


Figure 4.2: The probability of binarisation to +1 for each possible response to a Likert-7 scale survey question[1], given $\sigma \approx 0.2$.

where Φ is the standardised normal cumulative distribution function. This probability is small when x has large magnitude, or when σ is small. For the purposes of our experiments, we choose $\sigma = 0.203$ such that a 'weakly oppose' response to a 7-point Likert scale[1] (value $1/3$) is mapped to $+1$ with probability 0.05.

4.3 Parameter estimation

For the purposes of the experiments in the following sections, we perform parameter estimation to calibrate the belief system model described in Chapter 3.2 to the climate attitudes dataset (Chapter 9). We first outline the context of the parameter estimation problem, and then present our approach. We omit derivation details in this section; however, the curious reader can find these in Appendix A (**TODO**).

Consider a population of $M \in \mathbb{N}$ individuals with a shared belief system $\mathcal{M}$ comprising $N \in \mathbb{N}$ beliefs or attitudes with pre-defined adjacency matrix $\boldsymbol{A} \in \{0,1\}^{N \times N}$, for which the parameters (i.e., the baseline activations and interaction effects) are unknown. Suppose that for each individual $m \in [1, M]$ we have observed a series of measurements, reflecting that individual's belief system state at each of $t \in [1, T]$ uniformly spaced timesteps:

$$\left\{\boldsymbol{x}^t_{(m)}\right\}_{t=1}^{T}, \quad \text{where each } \boldsymbol{x}^t_{(m)} \in \mathbb{R}^N \tag{4.2}$$

The measurements need not be binary, but are assumed to be real. The collection of measurements across all individuals forms a dataset D, which is the particular observed value of the random variable $\mathcal{D}$ representing possible datasets.

[1]The oppose/support 7-point Likert scale has possible responses: strongly oppose, oppose, weakly oppose, neutral, weakly support, support, strongly support.

While maximum likelihood estimation (MLE) is often used to infer parameters for similar models (Lee et al., 2015; Nguyen et al., 2017), we should be cautious, for two reasons, of applying this method naïvely in the present study.

Firstly, the model defined in Equation 3.8 assumes binary spin states $s \in \{-1, +1\}$. The climate attitudes dataset does not satisfy this assumption and must be binarised for MLE to be applicable. However, we have no guarantee that the parameterisation inferred for any specific binarisation is a reasonable explanation for the non-binarised dataset, nor for any other possible binarisation in the case where a probabilistic binarisation scheme is used (such as the one described in Chapter 4.2).

Secondly, maximum likelihood estimation is prone to overfitting when the number of model parameters is similar to the number of observations (Epskamp, Borsboom, et al., 2018). While we consider only a small number of beliefs and attitudes in this study ($N \in \{7, 8\}$), the number of parameters p grows rapidly, with $p \in O(N^2)$ for both the symmetric and asymmetric model variants. In combination with sampling error, we expect the number of parameters to negatively impact the robustness of the inferred parameters — given an alternative dataset of the same size, we would expect significant variation in parameters.

We mitigate these respective issues in our parameter estimation scheme by (i) marginalising over the set of possible binarisations of D, and (ii) applying regularisation based on parameter magnitudes. We choose the parameterisation $\boldsymbol{\theta}^* \in \mathbb{R}^p$ which satisfies the optimisation problem:

$$\boldsymbol{\theta}^* = \underset{\hat{\boldsymbol{\theta}} \in \mathbb{R}^p}{\operatorname{argmax}} f\left(D; \hat{\boldsymbol{\theta}}\right), \quad \text{where} \quad f\left(D; \hat{\boldsymbol{\theta}}\right) \coloneqq \mathcal{L}_D\left(\hat{\boldsymbol{\theta}}\right) - \lambda \sum_{\theta \in \hat{\boldsymbol{\theta}}} \sqrt{\theta^2 + \varepsilon} \tag{4.3}$$

The first term in the objective function f, $\mathcal{L}_D\left(\hat{\boldsymbol{\theta}}\right)$, denotes the *expected* log-likelihood given a parameterisation $\hat{\boldsymbol{\theta}}$, over possible binarisations of D:

$$\mathcal{L}_D\left(\hat{\boldsymbol{\theta}}\right) = \mathbb{E}\left[L_{D_B}\left(\hat{\boldsymbol{\theta}}\right) \mid \mathcal{D} = D\right] = \sum_{D_B} P(\mathrm{Bin}(\mathcal{D}) = D_B \mid \mathcal{D} = D) \cdot L_{D_B}\left(\hat{\boldsymbol{\theta}}\right) \tag{4.4}$$

where $L_{D_B}\left(\hat{\boldsymbol{\theta}}\right)$ is the log-likelihood of a specific binarisation D_B. We will defer explicitly defining $L_{D_B}\left(\hat{\boldsymbol{\theta}}\right)$ for now, but will return to this point shortly in Equation 4.5.

TODO: Intuition for the effect of using the expectation.

- Inferred model reflects ambiguity in states close to zero.
- The likelihood contribution of each data value is the expectation over possible spin states.

- Values close to zero contribute a comparable weighting from each binary state.
- When the likelihood given one binary state would be significantly higher than for the other, this is downweighted to account for the probability that the value does not obtain this state.
- Therefore the expectation prevents overconfident optimisation toward any specific binarisation — the inferred model is required to explain the binarised dataset *in expectation.*
- In other words, if a data value may be binarised to -1 or $+1$ with equal probability, then both cases should be reasonably explained by the parameterisation.

The second term is a smooth variant of L1 regularisation (Tibshirani, 1996). This has the effect of penalising nonzero parameters which do not meaningfully contribute to the expected likelihood. Parameters instead must reflect relationships which are sufficiently prevalent in the data, reducing the robustness issues resulting from sampling error and a large number of parameters. The hyperparameter $\lambda \in \mathbb{R}^+$ controls regularisation strength, with higher values resulting in sparser models. As $\varepsilon \to 0^+$ the second term converges to the standard formulation of L1 regularisation. Unlike the standard formulation, when $\varepsilon > 0$ the first-derivative of this term exists, so is amenible to gradient-based optimisation. We discuss the choice of values for λ and ε in Chapter 4.3.1.

Let $D_B \sim \text{Bin}(D)$ be a possible binarisation of D, and $\boldsymbol{\theta}$ a parameterisation for the (asymmetric or symmetric) belief system model, which is decomposable into the model parameters $\boldsymbol{\theta} = \langle \boldsymbol{J}, \boldsymbol{h} \rangle$. The log-likelihood of D_B given $\boldsymbol{\theta}$ is:

$$\begin{aligned} L_{D_B}(\boldsymbol{\theta}) &= \log P(D_B \mid \boldsymbol{\theta}) \\ &= \sum_{m=1}^{M} \sum_{t=1}^{T-1} \log s_{i,(m)}^{t+1} \cdot h_i^{\text{eff}}\Big(\boldsymbol{s}_{(m)}^t\Big) - \log\Big(2 \cosh h_i^{\text{eff}}\Big(\boldsymbol{s}_{(m)}^t\Big)\Big) \end{aligned} \tag{4.5}$$

The derivation of Equation 4.5 is analogous to that of the non-equilibrium Ising model log-likelihood (Nguyen et al., 2017). Combining Equation 4.5 and Equation 4.4, we obtain an explicit expression for the expected likelihood:

$$\mathcal{L}_D(\boldsymbol{\theta}) = \sum_{\boldsymbol{s}} \left(\sum_{m=1}^{M} \sum_{t=1}^{T-1} P\Big(\boldsymbol{S}_{(m)}^t = \boldsymbol{s} \mid D\Big) \cdot \mathbb{E}\Big[\boldsymbol{S}_{(m)}^{t+1} \mid D\Big]^T h^{\text{eff}}(\boldsymbol{s}) \right) - Z(D; \boldsymbol{\theta}) \tag{4.6}$$

where $\mathbb{E}\Big[\boldsymbol{S}_{(m)}^{t+1} \mid D\Big]$ is the expected binary configuration for the observation $\boldsymbol{x}_{(m)}^{t+1}$, the vector $h^{\text{eff}}(\boldsymbol{s})$ contains the effective baseline activation for each spin given the previous state $\boldsymbol{s}$, and Z is the expected log partition function, summed across observations:

$$Z(D;\boldsymbol{\theta}) = \sum_{\boldsymbol{s}} \sum_{\substack{m \leq M \\ t < T}} P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) \sum_{i=1}^{N} \log\big(2\cosh h_i^{\text{eff}}(\boldsymbol{s})\big) \tag{4.7}$$

Per the optimisation problem defined in Equation 4.3, we choose the parameterisation $\boldsymbol{\theta}^*$ which maximises the regularised expected log-likelihood, which occurs when $\frac{\partial f}{\partial \theta} = 0$ for every parameter $\theta \in \boldsymbol{\theta}^*$. The partial derivative with respect to a given baseline activation parameter h_i, for $i \in [1, N]$, is given by:

$$\begin{aligned} \frac{\partial}{\partial h_i} f(D;\boldsymbol{\theta}) = &\sum_{\boldsymbol{s}} \sum_{\substack{m \leq M \\ t < T}} P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) \Big[\mathbb{E}\Big[S^{t+1}_{i,(m)} \mid D\Big] - \tanh\big(h_i^{\text{eff}}(\boldsymbol{s})\big)\Big] \\ &- \frac{\lambda h_i}{\sqrt{h_i^2 + \varepsilon}} \end{aligned} \tag{4.8}$$

Note that the expressions for $\mathcal{L}_D(\boldsymbol{\theta})$ and $\frac{\partial}{\partial h_i} f(D;\boldsymbol{\theta})$ are applicable to both the asymmetric and symmetric belief system models. This property does not hold for the partial derivatives with respect to interaction effects. In the asymmetric model, for $i, j \in [1, N]$, this is derived analogously to Equation 4.8 as:

$$\begin{aligned} \frac{\partial}{\partial J_{ji}} f(D;\boldsymbol{\theta}) = &\sum_{\boldsymbol{s}} \sum_{\substack{m \leq M \\ t < T}} P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) A_{ji} s_j \Big[\mathbb{E}\Big[S^{t+1}_{i,(m)} \mid D\Big] - \tanh\big(h_i^{\text{eff}}(\boldsymbol{s})\big)\Big] \\ &- \frac{\lambda J_{ji}}{\sqrt{J_{ji}^2 + \varepsilon}} \end{aligned} \tag{4.9}$$

In the symmetric belief system model, since we require that $\boldsymbol{J} = \boldsymbol{J}^T$, we estimate a single interaction parameter $J_{ji} = J_{ij} = \beta_{\{i,j\}}$ for each pair of spins S_i, S_j. Each pair thus contributes twice to the partial derivative, once for each direction:

$$\begin{aligned} \frac{\partial}{\partial \beta_{\{i,j\}}} f(D;\boldsymbol{\theta}) = &\sum_{\boldsymbol{s}} \sum_{\substack{m \leq M \\ t < T}} P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) \sum_{(i',j') \in \{i,j\}} A_{j'i'} s_{j'} \Big[\mathbb{E}\Big[S^{t+1}_{i',(m)} \mid D\Big] - \tanh\big(h_{i'}^{\text{eff}}(\boldsymbol{s})\big)\Big] \\ &- \frac{\lambda \beta_{\{i,j\}}}{\sqrt{\beta_{\{i,j\}}^2 + \varepsilon}} \end{aligned} \tag{4.10}$$

4.3.1 Hyperparameters

The optimisation problem in Equation 4.3 has two hyperparameters which must be specified prior to parameter estimation: the regularisation strength $\lambda \in \mathbb{R}^+$, and the smoothing parameter $\varepsilon \in \mathbb{R}^+$. The values for both are summarised in Table 4.1.

Parameter	Model type	Dataset	Value
λ	Symmetric	Full	1.4×10^{-2}
	Asymmetric	Full	1.2×10^{-2}
		Conservative	3.4×10^{-2}
		Liberal	4.3×10^{-2}
ε	All	All	10^{-8}

Table 4.1: Values for regularisation strength (λ) and smoothing (ε) hyperparameters. Regularisation strength model- and dataset-specific; *Conservative* and *Liberal* refer to subsets of the climate attitudes dataset comprising individuals with the specified ideology (Chapter 9).

We use the Extended Bayesian Information Criterion (EBIC) (Chen & Chen, 2008) to select λ, as recommended by Epskamp, Borsboom, et al. (2018):

$$\mathrm{EBIC}(\boldsymbol{\theta}^*) = k \cdot [\ln(M(T-1)) + 2\gamma \ln(p)] - 2\mathcal{L}_D(\boldsymbol{\theta}^*) \tag{4.11}$$

The variable k denotes the number of nonzero parameters in the optimised model, where a parameter θ is considered nonzero if $|\theta| > 10^{-2}$. We take $\gamma = 0.25$, which has been shown to work well in general for network inference in the inverse Ising problem (Barber & Drton, 2015).

The EBIC is an evaluative criterion for model selection, which balances maximisation of the log-likelihood with model complexity, as measured by the number of parameters. Compared to the original Bayesian Information Criterion, EBIC tends to be more conservative when the number of parameters and observatons are comparable (Foygel & Drton, 2010). We select λ which minimises the EBIC for the symmetric and asymmetric models (independently), the results of which are displayed in Figure 4.3.

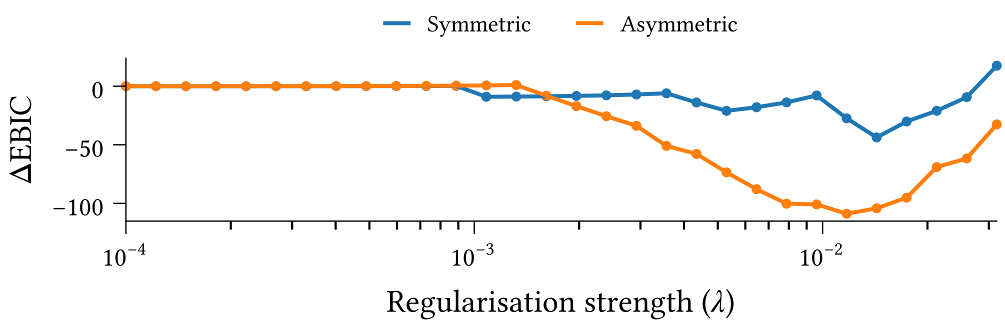


Figure 4.3: Effect of regularisation strength λ on Extended Bayesian Information Criterion for symmetric and asymmetric belief system models optimised to the climate attitudes dataset using Equation 4.3. The vertical axis measures the difference in EBIC compared to the no-regularisation case ($\lambda = 0$).

We consider parameters with magnitude less than 10^{-2} as 'effectively zero', and take $\varepsilon = 10^{-8}$ such that $\sqrt{\varepsilon}$ is significantly smaller than this threshold. This choice of ε ensures that the smooth regularisation remains a good approximation to L1 regularisation, i.e.,

$$\sqrt{\theta^2 + \varepsilon} \approx |\theta| \tag{4.12}$$

4.3.2 Implementation details

We solve the parameter estimation problem using the Scipy 1.17.1 implementation of the quasi-Newton BFGS optimisation algorithm (Nocedal & Wright, 2006, p. 136; Virtanen et al., 2020). We use the analytic jacobian comprising the partial derivatives stated above. We take $\boldsymbol{\theta} = \mathbf{0}$ as the initial guess.

To ensure numerical stability irrespective of dataset size, we rescale the expected log-likelihood contribution to the objective function and partial derivatives, dividing by the number of observed observations (time intervals) in the dataset. For $M \in \mathbb{N}$ individuals and $T \in \mathbb{N}$ timesteps this amounts to a scale factor of $\frac{1}{M(T-1)}$.

Chapter 5

Model Calibration

TODO:

- Discuss how all interaction terms are positive.
- Discuss differences in baseline activations.

Before proceeding, let's take a moment to calibrate and evaluate the models which will be used in the upcoming experiments. Using the parameter estimation method outlined in the previous chapter, we calibrate the symmetric and asymmetric belief system models to the climate beliefs dataset (see Chapter 9). We will evaluate the calibrated models on both structural accuracy ('how accurate are the parameter estimates?') and predictive capacity ('how well do the models explain the data?').

Figure 5.1 shows the interaction effect matrix, $\boldsymbol{J}$, for each model. We only display the upper triangular portion of the symmetric model's matrix, since each pair of spins in this model has a single, bidirectional relation. Both models feature a dominant diagonal, indicating that most variables are slow-moving with respect to the modelled timescale[2] (they are 'sticky'). This is particularly true for `Politics`, and less so for `CC Others Worry` and `CC Impact`.

As recommended in Epskamp, Borsboom, et al. (2018), we evaluate interaction parameter accuracy by examining the bootstrapped confidence intervals around each parameter estimate. We use bootstrapping to estimate the uncertainty in our parameter estimates due to sampling error. Let $\boldsymbol{D} \in \mathbb{R}^{M \times T \times N}$ be the complete dataset, where M, T, and N denote the number of participants, observations, and spins respectively[3]. We construct

[2] The models' timescales are set by the duration between survey responses, in this case approximately six months (Chapter 9).

[3] In the climate beliefs dataset we have $M = 1693$, $T = 2$, and $N = 8$

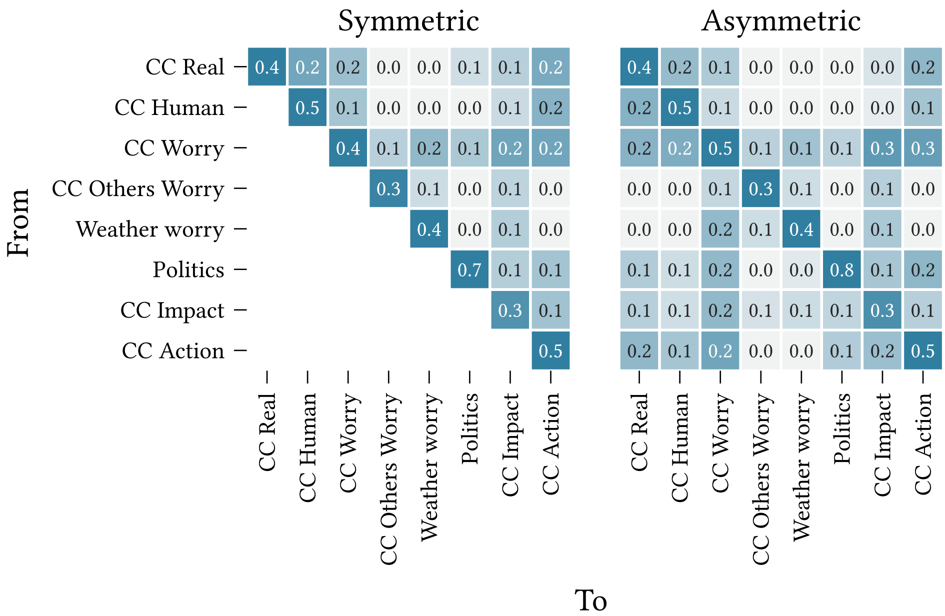


Figure 5.1: Interaction effect matrices ($\boldsymbol{J}$) for the symmetric and asymmetric belief system model variants, calibrated to the climate beliefs dataset (Chapter 9) using the parameter estimation method in Chapter 4.3.

500 bootstrapped datasets by sampling rows (participants) with replacement from $\boldsymbol{D}$, such that each bootstrapped dataset has the same shape as the complete dataset.

For each bootstrapped dataset $\boldsymbol{D}_{(i)}$ we calibrate a model $\mathcal{M}_{(i)}$ with parameters

$$\langle \boldsymbol{A}, \boldsymbol{J}_{(i)}, \boldsymbol{h}_{(i)} \rangle =: \boldsymbol{\theta}^*_{(i)} \in \mathbb{R}^p \tag{5.1}$$

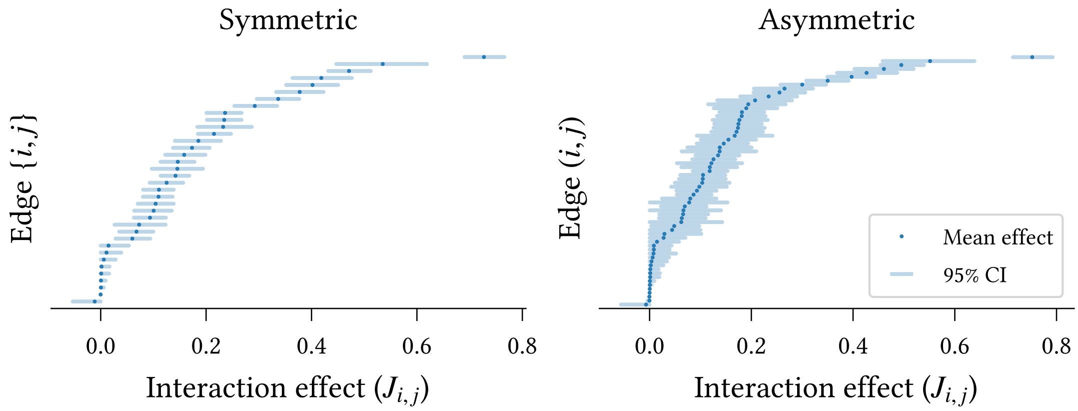


Figure 5.2: Interaction effect parameter accuracy for symmetric and asymmetric models calibrated to the climate beliefs dataset (Chapter 9). Mean parameter values are shown in increasing order. 95% confidence intervals are calculated using the percentile method over models calibrated to bootstrapped datasets (500 repeats).

where $p \in \mathbb{N}$ is the number of model parameters, and we take the adjacency matrix $\boldsymbol{A}$ as fully-connected, permitting influence relations to be inferred between each pair of beliefs or attitudes.

Figure 5.2 shows the estimated interaction effects in increasing order for each model. The confidence intervals are calculated using the percentile method (**CITE**) across the bootstrapped models' parameters. Most parameters display small 95% confidence intervals, indicating an accurate fit for both models. On average, the symmetric model exhibits smaller confidence intervals than the asymmetric model ($0.065 < 0.085$), which is expected given the larger number of parameters in the asymmetric model. In both models the self-interaction effect on `Politics` is significantly larger than all other parameters, as evidenced by the non-overlapping confidence intervals, supporting our earlier observation regarding Figure 5.1. Moreover, the remaining self-interaction effects are, in-general, significantly larger than the pairwise effects, with the exception of `CC Impact` in the symmetric model, and `CC Impact`, `CC Worry Others` in the asymmetric model.

Due to the use of regularisation in model calibration we must be careful not to draw conclusions regarding edge *existence* from this figure (Epskamp, Borsboom, et al., 2018). Instead, we may consider the proportion of bootstrapped models for which each edge is nonzero, shown in Figure 5.3. Relations which are selected in all models are not shown.

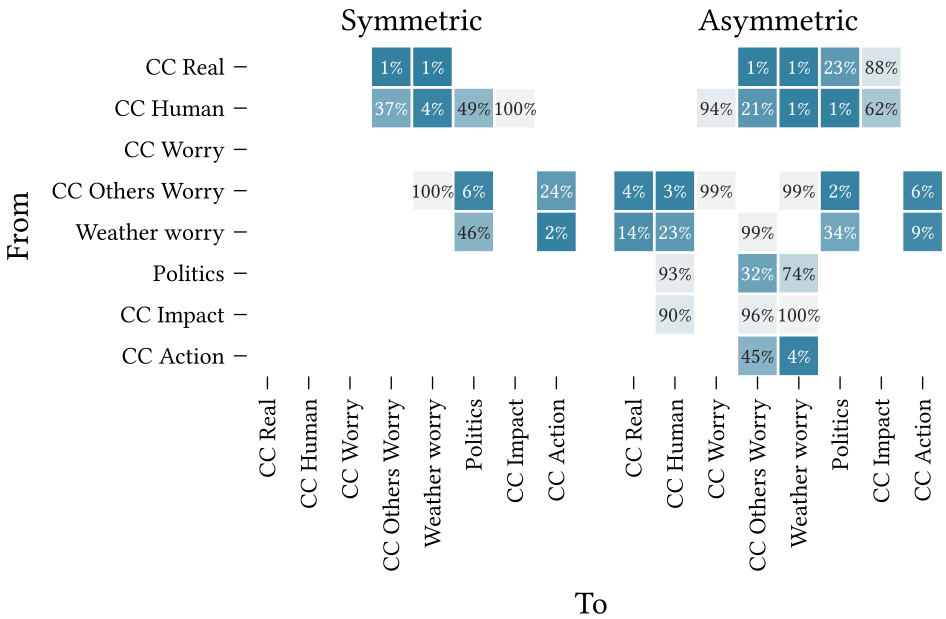


Figure 5.3: Edge selection probability for models calibrated to bootstrapped datasets sampled with replacement from the climate beliefs dataset (500 repeats). Edges are 'selected' if they survive regularisation with magnitude at least 0.01.

All edges excluded from the model calibrated on the full dataset (Figure 5.1) have low selection probability in the bootstrapped datasets ($< 50\%$). The relation `CC Real` $\to$ `Politics` is excluded from the complete asymmetric model but included (bidirectionally) in the symmetric model. This is reflected in the corresponding selection probabilities—this edge is selected in 100% of bootstrapped symmetric models but only 23% of asymmetric models. Edges which *are* selected in the complete models generally have high selection probability. The notable exception in the asymmetric case, `CC Human` $\to$ `CC Impact`, has a relatively small effect size ($J_{i,j} \approx 0.04$).

We now evaluate the models' predictive capacities[4]. First, we examine the reliability of predicted transitions. Figure 5.4 shows the average binarisation probability for each spin (i.e., the probability that the corresponding observation in the second wave of the dataset is binarised to +1), binned by transition probability. Bins with no observations are not shown (e.g., extreme values for `CC Others Worry`).

The two measurements are strongly correlated for most variables. The higher variation for `CC Real` and `CC Human` likely reflects the observed heavy skew in these variables toward larger values (see Figure 9.9 in Chapter 9). We observe no high/low probabilities

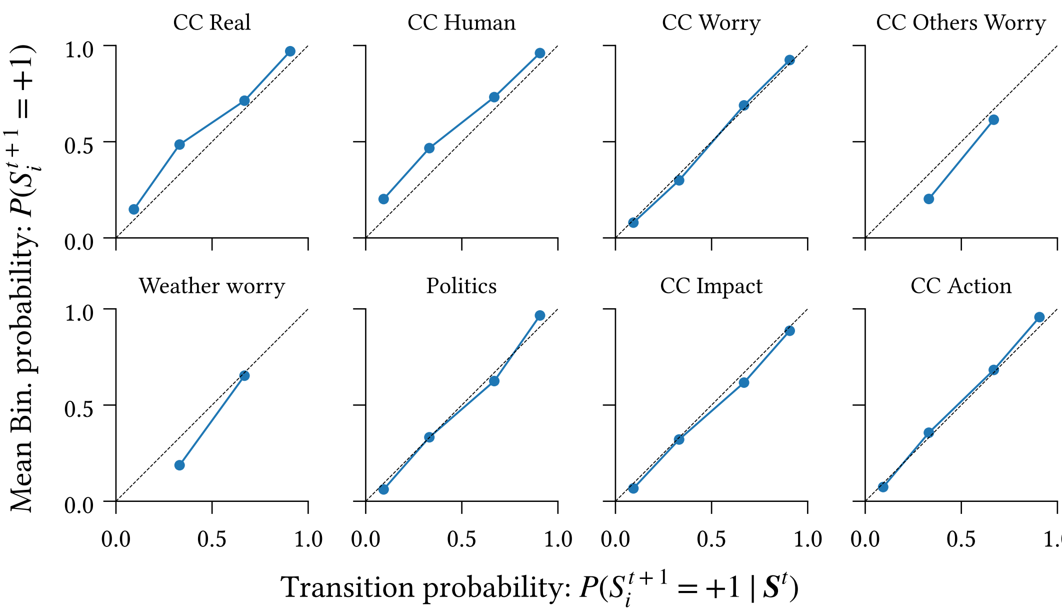


Figure 5.4: Reliability of transitions predicted by the model measured as the agreement between conditional probabilities under the model, $P(S_i^{t+1} = +1 \mid \boldsymbol{S}^t)$, and expected proportion of +1 values after binarisation, calculated as the average transition probability for survey participants in a given conditional probability bin.

[4]While we only show results for the asymmetric model here, the corresponding figures for the symmetric model are substantially similar, and all conclusions drawn regarding the asymmetric model apply also to the symmetric one.

for either `CC Others Worry` or `Weather Worry`, on account of the limited influence of other spins on these variables, as seen in the corresponding columns of Figure 5.1.

Next, we compare the calibrated model against a null model, in which we permit only baseline activation (h_i) and self-influence ($J_{i,i}$) parameters. We measure the relative entropy of the binarised data from the second wave of the dataset, with respect to the probability distributions induced by each model. For each participant and spin, we calculate the relative entropy as

$$D(P \parallel Q) \coloneqq H_C(P,Q) - H(P) \tag{5.2}$$

Where $P \coloneqq P\Big(\mathrm{Bin}\Big(X_{i,(m)}^{t+1}\Big)\Big)$ is the distribution over binarisations of the observation in the first timestep, and $Q \coloneqq P_{\mathcal{M}}\Big(S_{i,(m)}^{t+1}\Big)$ is the distribution over subsequent states according to the model, marginalising over binarisation of the previous observation,

$$Q \coloneqq P_{\mathcal{M}}\Big(S_{i,(m)}^{t+1}\Big) = P_{\mathcal{M}}\Big(S_{i,(m)}^{t+1} \mid \mathrm{Bin}\Big(X_{i,(m)}^{t}\Big)\Big) \cdot P\Big(\mathrm{Bin}\Big(X_{i,(m)}^{t}\Big)\Big) \tag{5.3}$$

The functions H and H_C denote the entropy and cross-entropy, respectively. The entropy term quantifies the uncertainty in the binarisation process, and the cross-entropy measures the 'expected surprise' when comparing predictions generated from the model with the true binarised dataset. Formally, these are defined as follows:

$$H(P) = -\sum_{s\in\pm 1} P(s)\log_2 P(s) \tag{5.4}$$

$$H_C(P,Q) = -\sum_{s\in\pm 1} P(s)\log_2 Q(s) \tag{5.5}$$

The relative entropy, $D(P \parallel Q)$, is small when either (i) the model accurately predicts the observed binarised value such that the cross-entropy is low[5], or (ii) the binarisation process is very uncertain. It follows that the relative entropy is high in cases where the model fails to accurately predict the next state, despite the binarisation process being fairly deterministic.

To test for the effect of including cross-interactions, we fit the fully-connected model and null models using 10-fold cross-validation. For each survey participant in the validation (holdout) split we calculate the mean difference in relative entropy between the fully-connected and null models. Figure 5.5 shows the mean relative entropy across individuals

[5]The relative entropy is strictly non-negative (**CITE**), so the binarisation entropy is a lower bound on the cross-entropy. This aligns with our intuition—the model cannot possibly be highly accurate when the binarisation process is very uncertain.

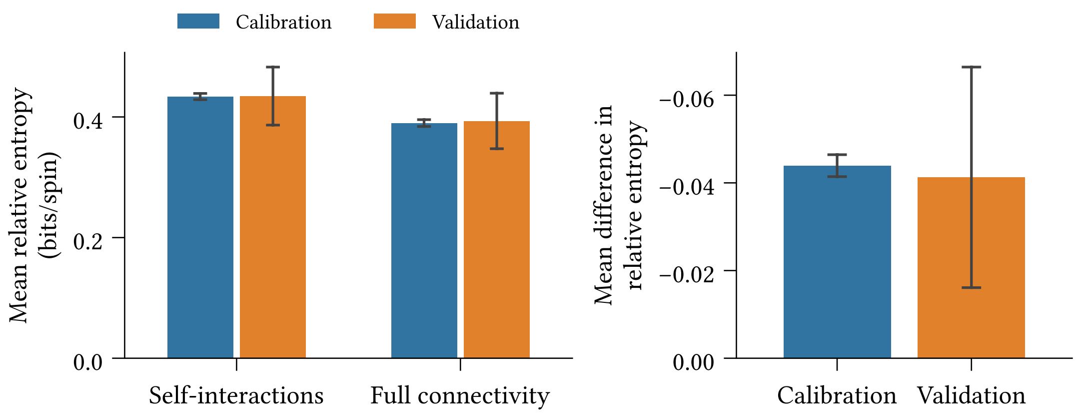


Figure 5.5: *(Left)* Mean absolute relative entropy of the binarised dataset with respect to the calibrated model, for the full-connected and null (only self-interactions) models, using 10-fold cross-validation. Average calculated across spins and survey participants. *(Right)* Mean difference in relative entropy. Confidence intervals display two standard deviations around the mean value.

and spins on the null and fully-connected models (left) and the mean *difference* in relative entropy, averaged across survey participants (right).

We find, in the right-hand panel, that the fully-connected model leads to significant reduction in relative entropy averaged over survey participants ($p < 0.05$). While statistically significant, the reduction is small. Since the measurement is averaged across survey participants, this finding suggests that the self-interaction model is sufficient to explain the behaviour for most participants—reflecting the slow-moving dynamics of the observed belief system—but not all participants. We anticipate that individuals whose belief states change more between observations are better explained by the fully-connected model than the null model.

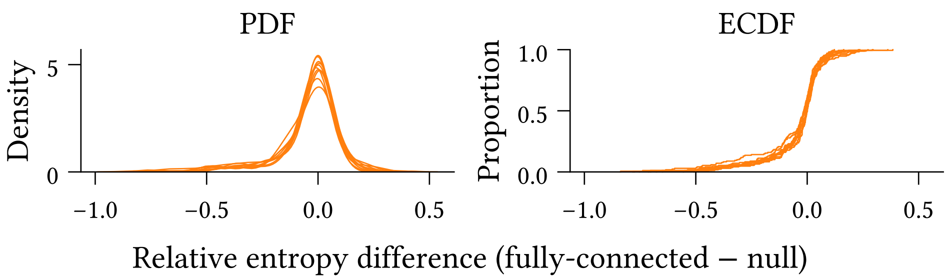


Figure 5.6: Probability density function (left) and empirical cumulative distribution function (right) of the mean difference in relative entropy between the fully-connected and null (only self-interaction) models across survey participants. Negative values indicate lower relative entropy for the fully-connected model.

Figure 5.6 explores this hypothesis, displaying the empirical probability density and cumulative distribution functions for the mean difference in relative entropy, across individuals. Indeed, we see substantial variation in effects. The weak right tail and heavy left tail indicate cases which are better-explained by the null and fully-connected models respectively.

We examine a sample of cases from either tail (top panel: left tail, bottom panel: right tail) in Figure 5.7. Each panel displays the change in binarisation probability between the two observed survey waves for a sampled survey participant. In the first three panels of the top row, we observe scenarios in which most spins are initially aligned, and a subset of the remaining spins then update to align with this set, i.e., where the system shifts toward a more consistent state. Conversely, in the first, second, and fourth panels in the bottom row we see the opposite scenario play out. That is, the system is initially well-aligned, yet some spins update to a less consistent state.

The observed behaviour aligns with our expectations, namely that the model with cross-interactions outperforms the null model in scenarios where the belief system state updates toward a more consistent state (according to the theory of cognitive dissonance), and is out-performed when participants' behaviour contradicts this theory.

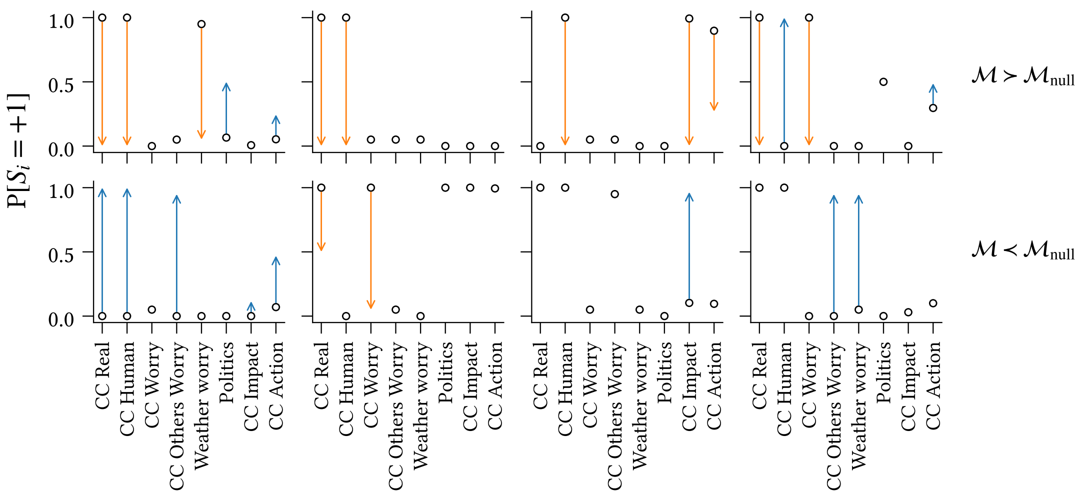


Figure 5.7: Observed transitions for sample survey participants in the left tail (top row) and right tail (bottom row) of Figure 5.6. Circles indicate the probability that participants' observations in the first wave are binarised to +1. Arrows denote the change in the second wave, colour-coded according to whether the change is toward the positive (blue) or negative (orange) state.

Chapter 6

Existence and impact of asymmetry in belief systems

Belief system models based on the Ising model, inspired by the cognitive dissonance theory of belief system dynamics, demonstrate impressive descriptive and explanatory capacities for various observed cognitive phenomena at both individual and collective scales. The currently prevalent view within the modelling community is that influence relations between beliefs or attitudes are symmetric — the pressure for X to align with Y is equal to that placed on Y to align with X. The present chapter places this assumption on trial.

We begin, in Chapter 6.1, by calibrating the non-equilibrium belief system model to the climate beliefs dataset described in Chapter 9.3. We demonstrate that belief system relations cannot be universally classified as symmetric, that some influence relations are recognisably asymmetric, and that relational asymmetry can vary in degree. In Chapter 6.2, we then show that this finding is not inconsequential. Asymmetric models exhibit different dynamics in experiments on belief-level interventions, with the potential to affect conclusions drawn about the relative effectiveness of interventions.

6.1 Existence of asymmetric relations

TODO:

- Consider adding squares to the asymmetric entries in Figure 6.2

To investigate the existence of asymmetric relations in the asymmetric model calibrated to the climate beliefs dataset, we examine the differences in directional interaction effects for the asymmetric models calibrated using bootstrapping in the previous section.

For each bootstrapped model, $\mathcal{M}_{(i)}$ with parameters $\langle \boldsymbol{A}, \boldsymbol{J}_{(i)}, \boldsymbol{h}_{(i)} \rangle$, we obtain an estimate for the directional differential matrix:

$$\Delta_J^{(i)} = \boldsymbol{J}_{(i)} - \boldsymbol{J}_{(i)}^T \tag{6.1}$$

Recall that the k'th row of $\boldsymbol{J}_{(i)}$ (for $k \in \{1, ..., N\}$) describes the strength and direction of influence *from* the spin S_k *toward* each other spin. Hence for $k, \ell \in \{1, ..., N\}$ we should interpret the element $\Delta_J^{(i)}|_{k,\ell}$ of the directional differential matrix as the excess influence of spin S_k on S_ℓ. A positive value indicates that S_k exerts greater influence on S_ℓ than S_ℓ does on S_k.

Figure 6.1 shows the median directional differential for each pair of spins in the asymmetric model, in decreasing order. The 90% confidence intervals are calculated as the 5th and 95th percentiles across bootstrap samples of the directional differential

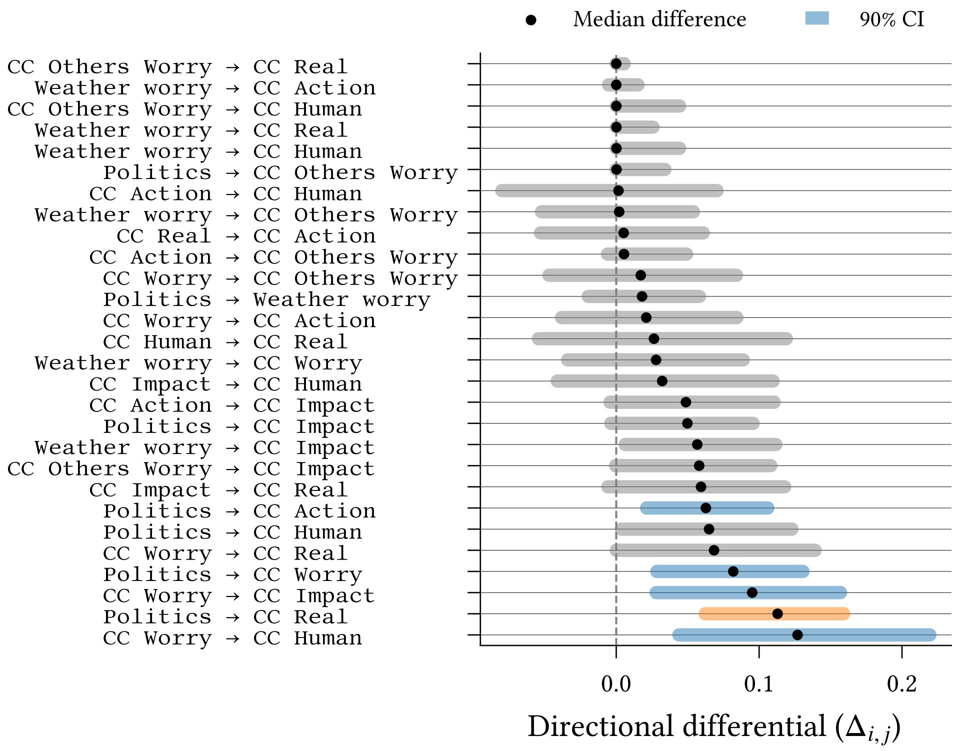


Figure 6.1: Directional differentials—the difference between directed interaction effects for a pair of spins—in reverse order of median value. Confidence intervals are calculated using the percentile method on models calibrated using bootstrapping ($n = 500$). Grey confidence intervals indicate absence of significant asymmetry. For cases with significant asymmetry, the colour indicates that one (**orange**) or both (**blue**) of the directed interacton effects are nonzero.

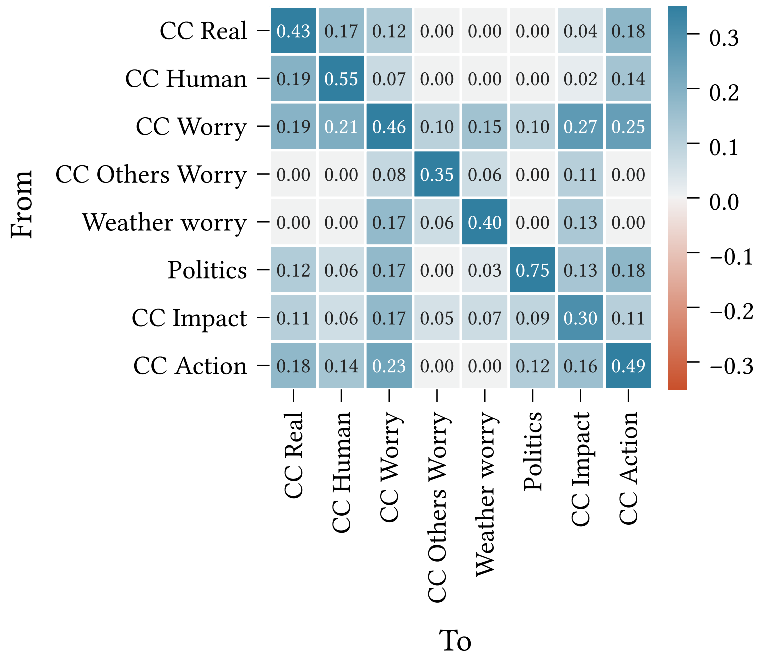


Figure 6.2: Interaction effect matrix ($\boldsymbol{J}$) for the asymmetric belief system model calibrated to the full climate beliefs dataset. This figure is identical to the asymmetric interaction matrix in the previous chapter (Figure 5.1).

matrix elements[6]. For each pair of spins we display only the directional differential for which the median is positive (since Δ_J is symmetric), and we exclude diagonal entries (which are zero by definition).

Figure 6.1 shows a majority of symmetric relations between spins (characterised by confidence intervals containing zero), with a small number of asymmetric relations. We can partition the asymmetric relations into two groups, according to whether directional interactions exist in only one direction, indicated using orange confidence intervals, or both directions with different strengths, indicated with blue confidence intervals (see Figure 6.2, identical to the interaction matrix shown in Figure 5.1 of the previous chapter). Pairs with two nonzero interactions comprise `Politics` $\rightarrow$ {`CC Action`, `CC Worry`} and `CC Worry` $\rightarrow$ {`CC Impact`, `CC Human`}; the only pair with a single directional interaction is `Politics` $\rightarrow$ `CC Real`.

Most directional differentials display substantial uncertainty (confidence interval width > 0.1 on average), reflecting the parameter accuracy discussed in the previous chapter (Figure 5.2). This results in several inconclusive cases, where the median bootstrap-estimate directional differential is nonzero but the confidence interval contains zero.

[6] In Chapter 5 we cautioned against using bootstrapped confidence intervals to test for the *existence* of edges by comparison with zero when regularisation is used during calibration—this caution does not apply to the *comparison* of edge weights via the mean difference (Epskamp, Borsboom, et al., 2018).

In other cases we observe apparent symmetric or near-symmetric relations (e.g., `Weather worry` $\longrightarrow$ `CC Others Worry`).

At the top of Figure 6.1 we see several apparently skewed confidence intervals. These all correspond to cases where both interactions are zero in the complete model (Figure 6.2), so the skews likely reflect differences in edge selection probability between the two directional interactions (Figure 5.3). For instance, `CC Others Worry` $\longrightarrow$ `CC Human` exhibits a larger apparent skew than `CC Others Worry` $\longrightarrow$ `CC Real`, and this is reflected in observed differences in selection probability.

Due to the large confidence intervals on most directional differentials we are less likely to observe significant asymmetric relations between variables with small interaction effects (since the possible difference in effects is smaller by definition). Indeed, all observed significant cases of asymmetry have at least one interaction effect with magnitude exceeding 0.1. This raises the question of whether asymmetry is instead simply explained by a high total interaction influence in combination with sampling error. However, this is disputed by `CC Action`, which has several strong interaction effects (its outbound effects are typically larger than those of `Politics`), yet does not exhibit significant asymmetric influence over any other variables. On the contrary, it is asymmetrically *influenced* by `Politics`.

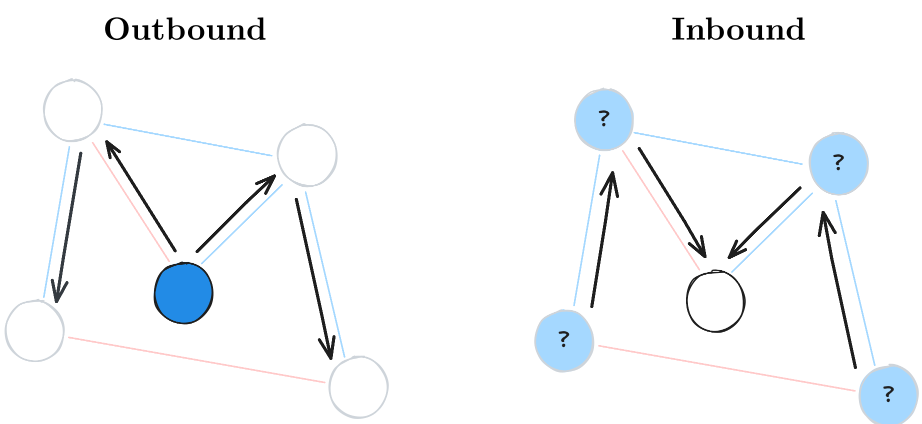
6.2 Asymmetry affects intervention dynamics

TODO:

- Discuss how the observed effects of intervening on politics are *in spite of* the high inertia on this variable. i.e., high inertia lowers the pre-intervention h_{eff}, making it harder to intervene (higher barrier to surpass).
- Comparatively, `CC Impact` is easier to intervene on (lower inertia), but ranks lower for the asymmetric model. Since all outbound interactions are similar (or better) than `Politics`, with the exception of `CC Action`, indicates that the added difficulty of overcoming the barrier on politics is outweighed by the greater influence on the target. Also perhaps due to the fact that the inertia works in favour of the intervention once politics flips.

We now investigate differences in belief system behaviour under intervention for symmetric and asymmetric models calibrated to the climate beliefs dataset. In this section we are concerned with *collective* impacts, e.g., the proportion of individuals whose behaviour shifts due to an intervention. This reflects population-level intervention scenarios, such as media campaigns or policies which affect most individuals. We will return to *individual* impacts of intervention in the following chapter (Chapter 7.1).

We consider both **outbound** and **inbound** interventions, illustrated below. Outbound intervention experiments (left) examine how interventions on a particular spin propagate to other beliefs and attitudes. On the other hand, inbound intervention experiments (right) consider a single belief or attitude as the intervention target, and examine the differences in effects on this spin when intervening elsewhere in the network. We refer to the belief or attitude on which an intervention is applied as the **point-of-intervention** and the belief or attitude whose resulting state is measured as the **target**.



For outbound intervention experiments we examine the effects of intervening on the following spins:

- `Weather worry`: Level of worry about future extreme weather events
- `CC Worry`: Level of worry about current and future climate change
- `Politics`: General political party identification and ideology

These choices are motivated by our findings in the previous section. Recall that `CC Worry` and `Politics` both feature in the set of asymmetric relations, including in the relation between these two variables. `Weather worry` does not appear in any significant asymmetric relations, and—in comparison with the other variables—has relatively small outgoing interaction effects.

For inbound intervention experiments we consider interventions seeking to influence the `CC Action` variable, which captures individuals' general support for pro-environmental action on climate change (e.g., specific policies, support for increased regulation, and affirmative attitudes toward individual responsibility on climate change).

We define two quantities of interest: the **effect of intervention**, and **effect of asymmetry**. These are used, respectively, to assess an intervention's impact on a given spin's behaviour with reference to our expectations in a no-intervention scenario (Definition 6.2.1), and the difference in a spin's behaviour in the asymmetric model, compared to our expectations in the symmetric model (Definition 6.2.2). Note that the effect of asymmetry is not inherently concerned with intervention, but is a general measure for the difference between asymmetric and symmetry model dynamics.

Both quantities compute a difference in effects between two distinct models: the intervention and null models for the effect of intervention; the asymmetric and symmetric models for the effect of asymmetry. To ensure outcome comparability, differences are computed between models with identical random number generation contexts.

Definition 6.2.1 (Effect of Intervention) Let $\mathcal{M}$ be a non-equilibrium belief system model, and denote the result of simulating the model for $t \in \mathbb{N}$ timesteps with initial state $\boldsymbol{s}_0$ as $\mathcal{M}^t(\boldsymbol{s}_0)$.

For an intervention model $\mathcal{M}_\delta$ with an arbitrary point-of-intervention, we define the **effect of intervention** for $\mathcal{M}_\delta$ as the change in outcome at time $t \in \mathbb{N}$, with respect to the null (no-intervention) model, $\mathcal{M}_0$:

$$f_{\text{int}}(\mathcal{M}_\delta) = \mathcal{M}_\delta^t(\boldsymbol{s}_0) - \mathcal{M}_0^t(\boldsymbol{s}_0) \tag{6.2}$$

Definition 6.2.2 (Effect of Asymmetry) Let $\mathcal{M}_{\text{asym}}$ be a calibrated asymmetric non-equilibrium belief system model with an arbitrary intervention applied, and $\mathcal{M}_{\text{sym}}$ be a corresponding symmetric model calibrated to the same dataset. We define the **effect of asymmetry** for $\mathcal{M}_{\text{asym}}$ as the difference in the effect of intervention with respect to the symmetric model:

$$f_{\text{asym}}(\mathcal{M}_{\text{asym}}) = f_{\text{int}}(\mathcal{M}_{\text{asym}}) - f_{\text{int}}(\mathcal{M}_{\text{sym}}) \tag{6.3}$$

In each of the intervention experiments described below, we use the models calibrated to the full climate beliefs dataset (Chapter 5). For a given experiment, we initialise the associated model (symmetric/asymmetric; intervention/null) with the binarised observations from the final wave of the climate beliefs dataset. We then draw consecutive samples from the model using parallel glauber dynamics (Chapter 3.4) until $t = 5$, before measuring the state of the target spin. Since the duration between waves in the climate beliefs dataset is approximately six months, the choice of $t = 5$ corresponds to measuring the impact of intervention after roughly two and a half years in the model timescale. To account for stochasticity in both the binarisation and simulation procedures we perform 500 repeats of each intervention.

We first consider the impact of intervention strength on effect of intervention. The strength of an intervention affects the degree to which the intervention changes the state of the point-of-intervention. However, this is also sensitive to the existing influence on that spin[7]. Since the climate beliefs dataset comprises individuals with different states, it follows that intervention strength may impact not only the degree to which the behaviour of a target spin changes, but also the relative effects of different interventions.

To understand the potential impacts of intervention strength, we compare the mean effect of intervention across individuals for intervention strengths $\delta_h \in \{0.5, 2.5\}$, for each of the outbound points of intervention listed above Figure 6.4. The choice of intervention strengths is motivated by our earlier analysis in Chapter 3.5, with the two strengths corresponding to 'weak' and 'strong' interventions respectively, as judged by their direct impact on the behaviour of the point-of-intervention.

We observe a strong linear relationship between intervention strength effects for each point-of-intervention, indicating that—at the population-mean level—the impact of intervention strength on effect is mostly one of scale, that applies similarly to all target spins. For the purposes of the following experiments it then suffices to consider only a single intervention strength. As our current focus is on intervention propagation and the effects felt at spins other than the point-of-intervention, we take $\delta_h = 2.5$, such that

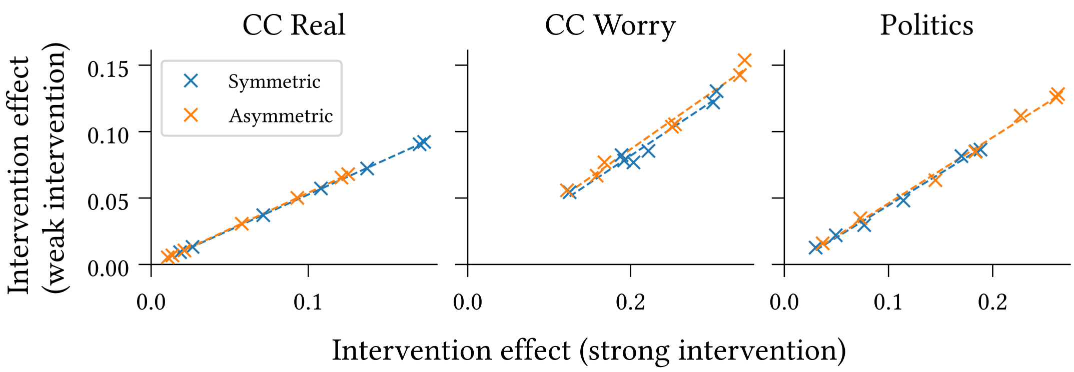


Figure 6.4: Weak ($\delta_h = 0.5$) and strong ($\delta_h = 2.5$) interventions exhibit strong positively correlated effects of intervention, consistently across points-of-intervention (panels) and intervention targets (scatterplots). This finding holds for both symmetric and asymmetric models calibrated to the climate beliefs dataset.

[7]For detailed discussion on the selection and implications of intervention strengths see Chapter 3.5.

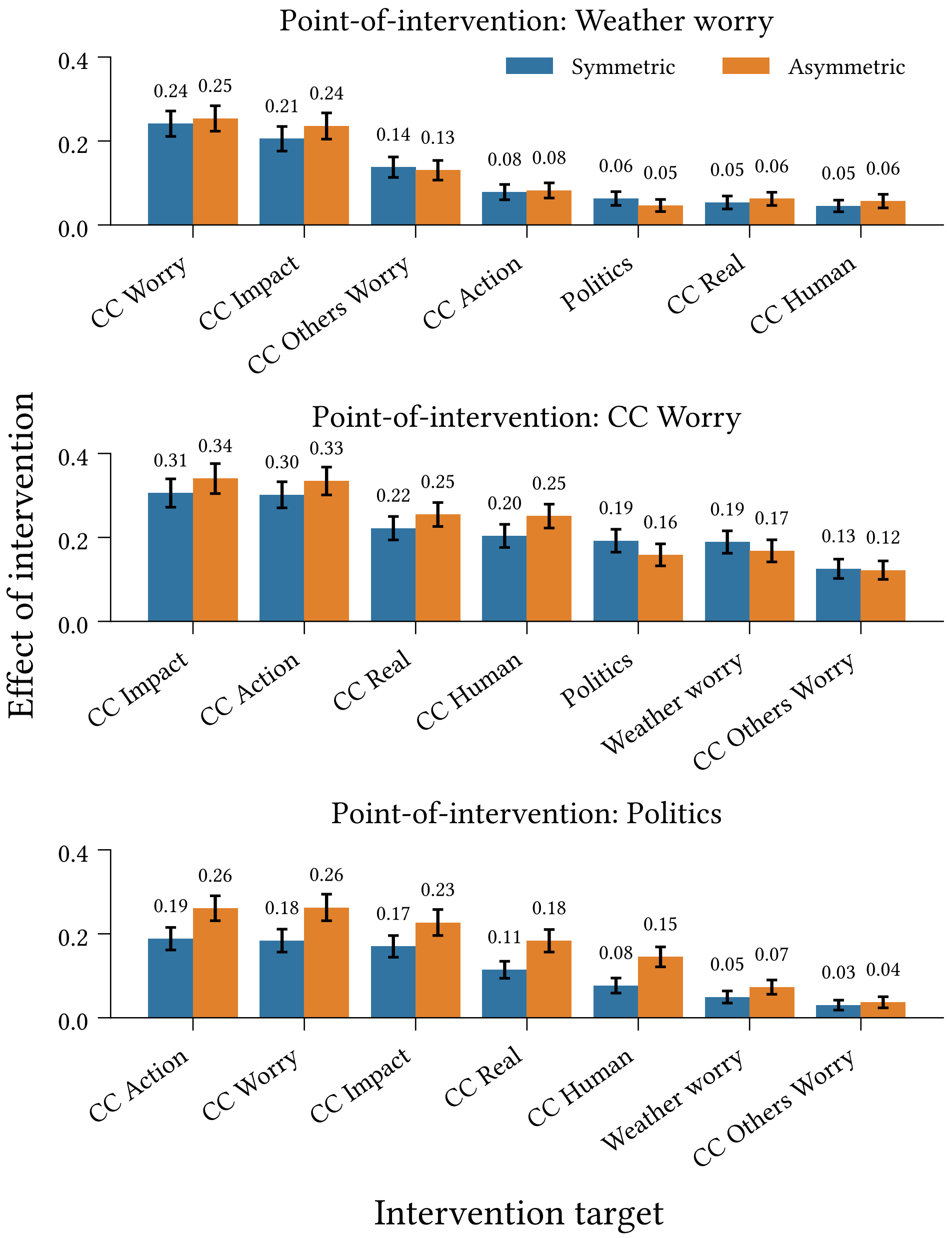


Figure 6.5: Outbound effect of intervention (Definition 6.2.1) at each target spin for interventions targeting `Weather Worry`, `CC Worry`, and `Politics`, in symmetric and asymmetric belief system models calibrated to the climate beliefs dataset. Intervention strength:w $\delta_h = 2.5$. Confidence intervals display 1.96 standard deviations around the mean effect, measured across 500 repeated simulations.

the interventions can be considered generally effective at shifting the behaviour at the point-of-intervention.

Figures 6.5 and 6.6 show the outbound mean effects of intervention and asymmetry, respectively, across individuals, for the points of intervention listed above. The confidence intervals show two standard deviations around the mean values.

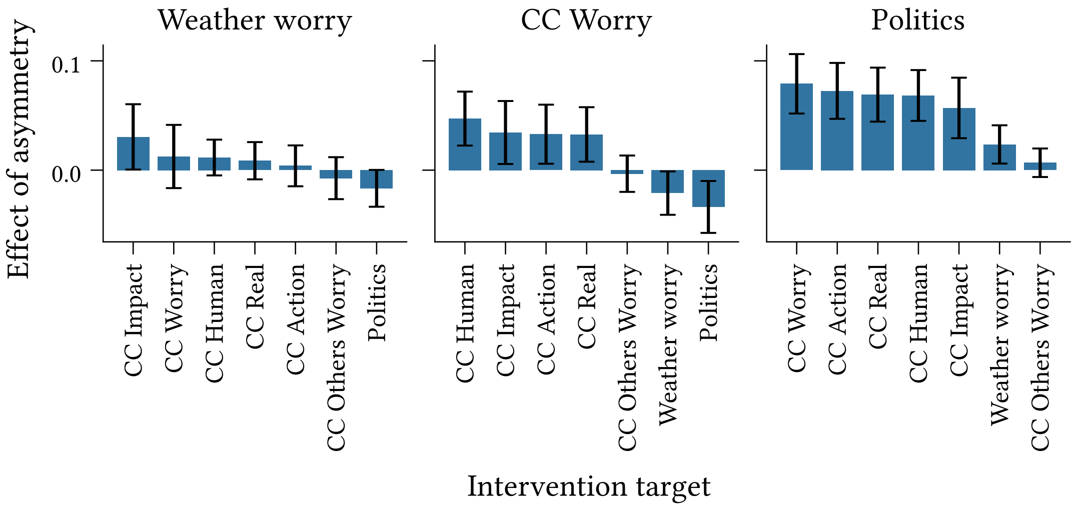


Figure 6.6: Outbound effect of asymmetry (Definition 6.2.2) at each target spin for interventions on `Weather Worry`, `CC Worry`, and `Politics`, for models calibrated to the climate beliefs dataset. Intervention strength: $\delta_h = 2.5$. Confidence intervals display 1.96 standard deviations around the mean effect of asymmetry ($n = 500$).

We observe, in Figure 6.5, that different targets exhibit different effects of intervention. The effect generally decreases with the magnitude of the interaction from the point-of-intervention to the target spin; however, we also note that all effects are positive. This suggests that for the measurement time examined here, interventions act primarily through direct interactions with the target spin, with some indirect influence propagating through intermediary interactions. For instance, while `Weather Worry` has no direct influence on `CC Action`, we see nonzero effect of intervention as a result of indirect paths, e.g., through `CC Worry`. We see further evidence of this through comparison with an analogous figure (see Figure B.1 in Appendix) using a measurement time of $t = 10$ (approximate five years in the models' timescales), which shows larger increases in the effect of intervention for variables with stronger incoming interactions from non-intervention spins.

While the confidence intervals for the symmetric and asymmetric models overlap significantly for most targets in Figure 6.5 (with some exception in the bottom panel), the effects of asymmetry displayed in Figure 6.6 provide a cleaner comparison. We observe no significant differences between the two models for the `Weather Worry` scenario. For the remaining two scenarios, however, we see several significant differences. In particular, all pairs of variables with asymmetric direct relations—identified in the previous section—display significant differences.

Comparing `Politics` and `CC Worry`, we observe that while `CC Worry` exhibits greater effects of intervention (Figure 6.5), `Politics` exhibits larger mean effects of asymmetry

(Figure 6.6), though the latter differences are not statistically significant. In both cases we see that in the symmetric model the strengths of cross-interactions with other spins are typically between the corresponding inbound and outbound asymmetric interaction strengths. We contrast this with `CC Impact`, which has mostly symmetric relations. While `CC Impact` and `Politics` have extremely similar outbound interaction strengths in the asymmetric model, `CC Impact` is considerably more influential than `Politics` in the symmetric model (i.e., it has higher average interaction effects).

Moreover, while `CC Worry` has inbound and outbound interactions with all other spins in both models, `Politics` has different inbound and outbound connectivity in the asymmetric model. We see this reflected in the symmetric model, which omits interactions with `CC Human` and `Weather Worry` entirely despite, outbound interactions to these spins in the asymmetric model. This explains the observed differences in the effects of intervention and asymmetry: `CC Worry` has higher effects of intervention in both models on account of its strong outbound interaction effects, which are also mostly retained in the symmetric model, while the removal of interactions for `Politics` results in lower influence and fewer (direct and indirect) paths in the symmetric model, compared with the asymmetric model.

Turning our attention toward inbound intervention dynamics, we now consider the effects of different interventions seeking to influence the state of `CC Action`. The top panel of Figure 6.7 shows the mean effect of intervention observed at the target spin for each possible point-of-intervention. Examining the mean effects allows us to gauge the expected absolute effect of each intervention, but is limited for purposes of comparing the relative effects of interventions.

The mean effect of intervention, calculated separately for each point-of-intervention, erases information regarding the expected relative effectiveness of different interventions. To assess relative effectiveness, we therefore also estimate the expected ranking over points-of-intervention (the bottom panel of Figure 6.7). For each repeat, we order the points-of-intervention in increasing order of average collective effect of intervention. Higher values are assigned higher ranks. In the case of a tie, we assign all tied spins the minimum rank which would have been assigned to the group, had they been distinct[8].

As with the outbound case, we find that the effect of intervention on `CC Action` varies depending on where we intervene. Points-of-intervention (along the x-axes) are sorted in decreasing order of mean values. We observe identical orderings between the effect

[8]For instance, suppose we have five variables, $\{A, B, C, D, E\}$ with average collective effects $\{0.2, 0.2, 0.6, 0.4, 0.1\}$. Then the assigned ranking would be $\mathrm{Rank}(A, B, C, D, E) = [2, 2, 5, 4, 1]$. The variables B and C are both assigned the rank 2. No variable is assigned rank 3.

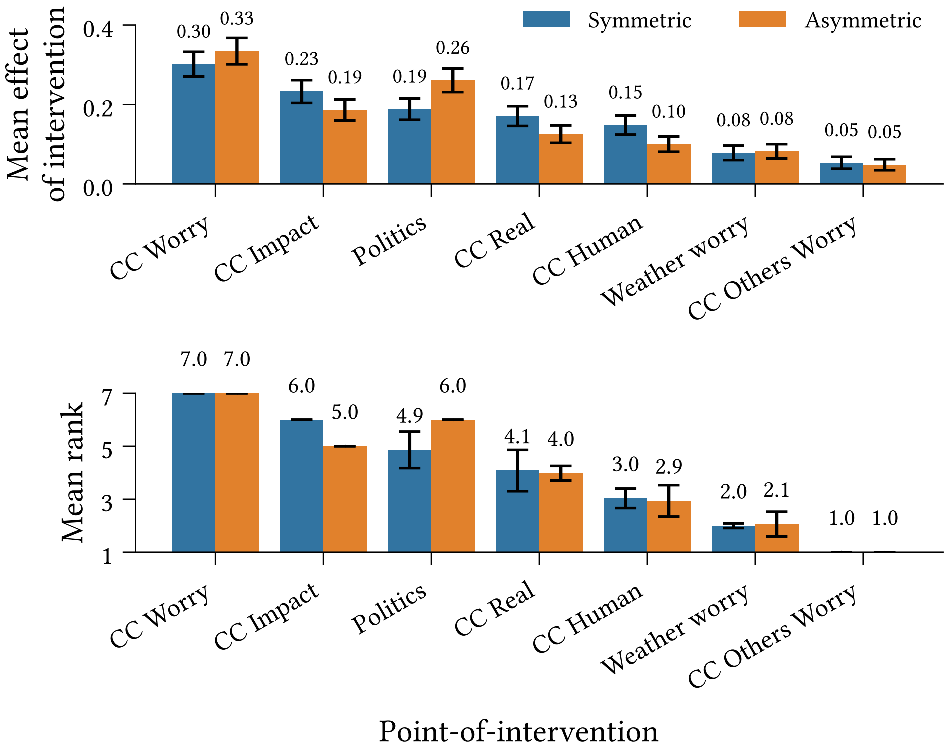


Figure 6.7: Inbound mean effect of intervention (Definition 6.2.1) (*Top*) and mean rank (*Bottom*) for interventions targeting `CC Action`. Ranks are calculated per-repeat with respect to the average effect of intervention—higher values are assigned higher ranks. Confidence intervals display 1.96 standard deviations ($n = 500$).

of intervention and ranking plots, showing good correspondence between the two measures. We observe a clear inversion in this ordering for the asymmetric model, with `Politics` exhibiting the second-highest rank, and (at least) second-highest mean effect of intervention ($p < 0.05$).

Unlike with the outbound effects, the order of effect sizes does not align well with the corresponding strengths of direct interactions strengths into the target variable. For instance, `CC Real` and `CC Human` have the second- and third-strongest cross-interactions into `CC Action` in the symmetric model, yet the fourth- and fifth-highest effect of intervention. This likely reflects the low outbound connectivity of these variables in comparison with `CC Impact` and `Politics`, despite the latter spins having relatively lower direct on the target—additional evidence that indirect influence plays a critical role in intervention dynamics.

Most confidence intervals around the mean ranks are small, indicating negligible (if any) variation between simulations or possible binarisations of the dataset. However, we do observe three larger, overlapping confidence intervals for each model. These arise due to variation the mean effect of intervention causing occasional inversions of the ranking order. Note that we only observe inversions between points-of-intervention which are adjacent in the sorted order.

Chapter 7

Heterogeneous belief systems and intervention effects

In this chapter we investigate individual heterogeneity in both intervention behaviour and belief system structure in asymmetric, non-equilibrium belief system models.

While the previous chapter was primarily concerned with the *collective* effects of intervention, the distribution of expected effect of intervention across individuals for inbound interventions targeting ‘Climate Action’ shows substantial variation between survey participants (Figure 7.1). We begin this chapter with an investigation into the conditions under which different interventions targeting attitudes on climate action are likely to be effective.

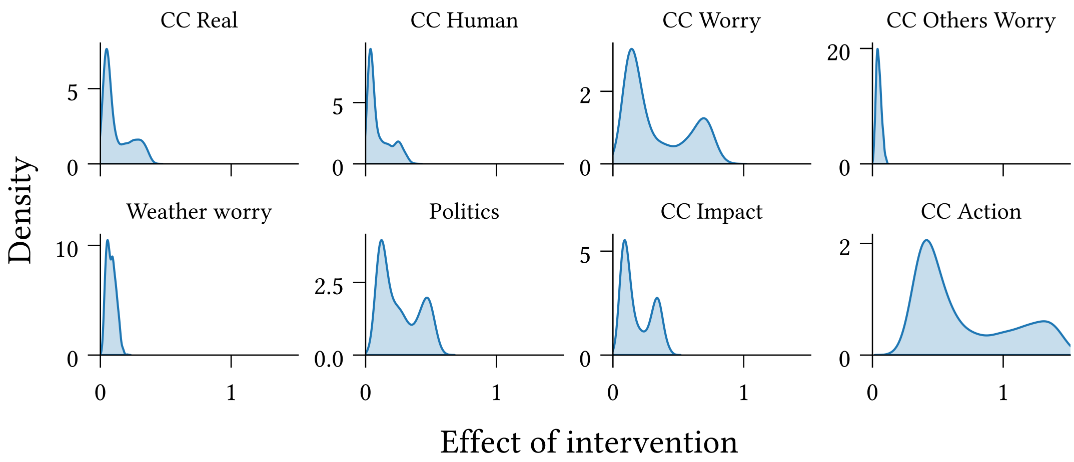


Figure 7.1: The expected effect of intervention for strong interventions ($\delta_h = 2.5$) targeting ‘Climate Action’ exhibits substantial variation between individuals.

We then consider how inferred belief system structure varies between individuals, or subgroups in a population. In the previous chapter we identified political ideology as a particularly influential attitude dimension, both through asymmetric interactions with other variables (Chapter 6.1) and as an effective point-of-intervention for interventions targeting attitudes on climate action (Chapter 6.2). Here, we investigate whether this influence extends beyond pairwise interactions with other variables by comparing asymmetric belief systems calibrated to the (self-reported) liberal and conservative subpopulations within the climate beliefs dataset.

7.1 Heterogeneity in intervention effects

In the intervention simulations described in the previous chapter, the only distinguishing factor between survey participants is their binarised initial state derived from the final wave of the climate beliefs dataset. It follows that any difference in belief system dynamics between distinct individuals—after accounting for simulation stochasticity—is the result of their different initial states. To identify the conditions under which an intervention is effective, it then suffices to characterise the set of *initial states* which yield effective interventions, and distinguish them from those that do not.

Our goal is to find a function which maps from an intervention effect to sets of (unbinarised) states that yield that effect, $g : \mathbb{R} \to 2^X$, where $X = [-1, +1]^N \subset \mathbb{R}^N$. We'll call g the **effect characterisation function**.

Consider a function which does the opposite, mapping initial states to a measure of intervention effect, $f : X \to \mathbb{R}$. Given such a function, we can straightforwardly construct the corresponding effect characterisation function as:

$$g : y \mapsto \left\{x \in \pm 1^N \subset \mathbb{R}^N \mid f(x) = y\right\} \tag{7.1}$$

However, while technically satisfying the definition of the effect characterisation function, such a function would be useless for qualitatively determining the kinds of initial states which yield effective interventions. The elements of the codomain have (potentially) infinite cardinality, and are not immediately interpretable. For our purposes, we are less interested in the *specific* states that yield a given intervention effect, but more so in a *concise description* of that set of states.

Regression decision tree models can provide such descriptions, using inequality bounds to partition the initial state space into regions, each of which is assigned a predicted effect. Given a parameterised regression decision tree, we may construct a concise effect

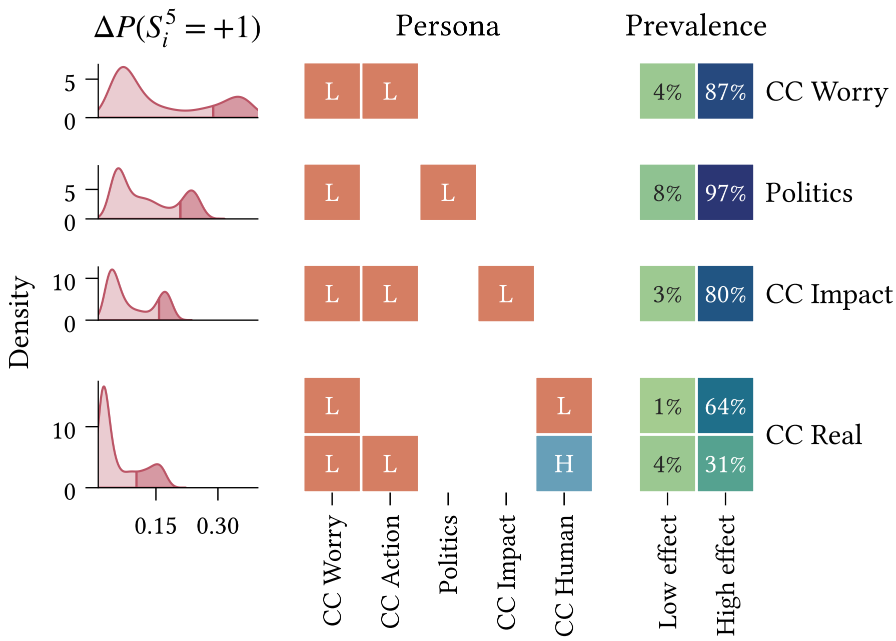


Figure 7.2: Characterisation of initial states ('personas'; middle panel) which yield effective interventions when targeting `CC Action` using different points-of-intervention (rows). Intervention effect is measured as the difference in `CC Action` activation probability between intervention ($\delta_h = 2.5$) and null (no-intervention) models at $t = 5$, for the asymmetric belief model calibrated to the climate beliefs dataset. Differences in the population upper quartile are 'high-effect'. (**Left**) Distribution of intervention effect across survey participants. (**Centre**) Typical high-effect personas estimated using a depth-3 regression decision tree. Rows are distinct personas. Cells correspond to intervals of initial state dimensions ($L = [-1, 0]$, $H = [0, +1]$). (**Right**) Prevalence of each persona in the high-effect and low-effect subpopulations.

characterisation function by identifying, for each predicted effect, the combination of inequalities which define the corresponding infinite set of initial states. When the parameter estimation algorithm is restricted to shallow trees (e.g., depth 3 or 4, referring to the number of inequality bounds defining each region), these combinations can also be interpretable as rules or *personas*.

Figure 7.2 shows the effect characterisation functions for different interventions targeting `CC Action`, obtained from shallow regression decision trees with depth 3. The response variable is the expected difference between the activation probabilities for `CC Action` at $t = 5$, in the intervention and null models, estimated per-individual using the average probabilities across 500 repeated simulations. Each model is calibrated to the full climate beliefs dataset (see Chapter 5 for calibration details). As in the previous chapter, we use a fixed intervention strength of $\delta_h = 2.5$. The decision trees are fit using a mean-squared error objective function, and each achieve R^2 values of at least 0.9 (estimated using 10-fold cross-validation).

Each row in the figure corresponds to a different point-of-intervention (specified on the right). The left-hand column shows the distribution of differences in activation probabilities across individuals. The central column shows the personas which yield intervention effects within the upper quartile. Each row is a separate persona, and the columns refer to different dimensions (beliefs/attitudes) of the initial state. The personas themselves are specified using a compressed representation[9]: L $\mapsto [-1, 0]$ and H $\mapsto [0, +1]$. The right-hand column displays the prevalence of each persona within/outside the upper quartile of individuals, showing the proportional make-up of the high-effect population. We exclude points-of-intervention with no sufficiently high-effect observations (e.g., `CC Others Worry` in Figure 7.1), classified as an upper quartile less than 0.1 (`CC Human`, `CC Others Worry`, `Weather Worry`). We only include personas with at least 15% prevalence in the observed upper quartile.

The expected difference in activation probabilities between the intervention and null models exhibits a clear bimodal distribution across survey participants for each scenario, with the higher mode contained within the upper quartile (indicated by the darker shaded regions). For each point-of-intervention we observe a small set of personas. In each case these exhibit high prevelence among individuals with high intervention effects, and considerably lower prevalence for other individuals, indicating that the identified personas effectively characterise the conditions for effective interventions[10] .

For the present analysis, it is important to recognise that the intervention effect measured here does not, in general, reflect the resulting change in behaviour with respect to the initial state. It is entirely possible that both intervention and null models lead to an increase, or decrease, in the probability of desired behaviour for the target spin. Hence it is most appropriate to view 'high-effect' interventions as those which achieve substantially more desirable states than would be observed given no intervention.

With this in mind, we now consider several specific features of interest in the identified personas. Firstly, we observe that all personas require a low initial state for `CC Worry`. This may result from this variable's low inertia and high connectivity — in particular, its large outbound interaction effect toward the target variable (see Figure 6.2). These factors result in `CC Worry` being relatively influential and influentia*ble*, and therefore an effective indirect pathway for various interventions targeting `CC Action`. The significance

[9]As our intention is to capture coarse patterns, we round the original `f64` values describing the interval boundaries extracted from the model to the nearest multiple of 0.5.

[10]Note that the personas are not necessarily complete. For instance, suppose that a pair of variables are highly correlated in the initial state, and are Low whenever the intervention effect is high. A complete characterisation includes both variables; however, a decision tree is likely to include only one, since after splitting on one of the two variables, the other is redundant. This is especially true for shallow decision trees.

of the requirement that `CC Worry` be *low* is evident when comparing the implications for the null and intervention models. Due to `CC Worry`'s considerable outbound interactions, an intervention which successfully activates this variable has increased potential for propagation. In the null model, however, these interactions work against the desired result—if `CC Worry` remains low, it exerts this influence on all other spins.

Second, we observe that for each point-of-intervention (with the exception of `CC Real` on account of high correlation with `CC Human`[10]), a necessary condition for high effect is that the initial state of point of intervention itself be low. That is, for an intervention on X to be successful, X must not already be too high. This aligns with our prior expectations regarding the varied effects of interventions with respect to pre-intervention state (see Chapter 3.5).

Finally, the `CC Real` scenario is unique in that it includes two prevalent personas. The more prevalent persona corresponds to situations in which individuals are concerned about the current/future effects of climate change, but do not believe that climate change is human-caused[11]. In the case where an individual does believe that climate change is human-caused, this intervention may still be effective, so long as they do not already have a positive attitude toward climate action.

7.2 Heterogeneity in belief system structure

To-do:

- Include directional differential figure per-ideology (as in Figure 6.1).
- Discuss differences in baseline activations.

Until this point, we have considered belief systems as common to a population of individuals. However, the relations between beliefs and attitudes are inherently individual in nature. The existence, direction, and degree of relation between two beliefs is dependent on an individual's own beliefs regarding their relatedness.

Here we investigate the extent to which belief systems may vary between groups of individuals with different self-reported political ideologies. While this is still far from representative of the differences between individuals (Brandt & Morgan, 2022), it will allow us to examine differences in general trends for these subpopulations. Using the parameter estimation approach described in Chapters 4.3 and Chapter 5, we calibrate separate asymmetric belief systems to the subsets of the climate beliefs dataset which

[11]The negative state for `CC Human` is an aggregation of the beliefs that climate change is a natural phenomenon, and that climate change is not real (has no causes). See Chapter 9 for further details.

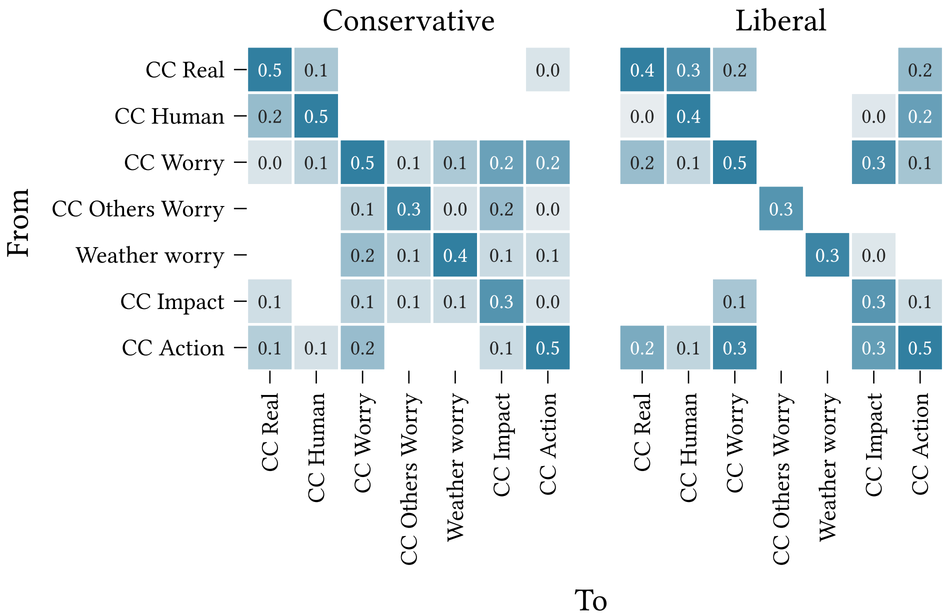


Figure 7.3: Interaction effect matrices ($\boldsymbol{J}$) for asymmetric belief systems calibrated to the conservative and liberal subsets of the climate beliefs dataset (Chapter 9) using the parameter estimation method in Chapter 4.3.

consistently (i.e., in both waves) self-report conservative ($n = 507$) and liberal ($n = 375$) political ideologies[12]. We exclude the `Politics` variable from the model, since this is captured by the partitioned datasets. The hyperparameters used for regularisation strength and smoothing are listed in Table 4.1.

Figure 7.3 shows the interaction effect matrices, $\boldsymbol{J}$, for the conservative and liberal subpopulations. We observe several differences between the two models, as well as with comparison to the model calibrated on the full dataset (Figure 6.2). Notably, the model calibrated to the liberal subpopulation has higher sparsity (proportion of missing cross-interaction edges) and mean interaction effect over cross-interactions than either the conservative model or the complete model (Table 7.1). Among the interactions present in the liberal model, 82% also occur in the conservative model, contrasting the 54% of conservative interactions which are also in the liberal model.

The conservative model displays broad (yet mostly weak) reinforcing interactions between climate-related concerns, beliefs and others' concerns, and climate-impact beliefs. This contrasts the liberal model, in which only `CC Worry` and `CC Impact` are non-trivially related. Moreover, we observe that `CC Others Worry` and `Weather Worry` in fact have *no* incoming cross-interactions.

[12]Note that the total number of samples across the conservative and liberal subsets is smaller than the number of samples in the complete climate beliefs dataset, since we only retain data from participants whose ideology is with consistent across survey waves.

Data subset	Sparsity	Mean interaction
Full	0.30	0.13
Conservative	0.38	0.10
Liberal	0.60	0.17

Table 7.1: Sparsity (proportion of missing cross-interactions) and mean interaction effect (over cross-interactions) for asymmetric models calibrated to the conservative and liberal subsets of the climate beliefs dataset, compared with the model calibrated on the complete dataset.

We also note differences in the interaction effects influencing `CC Action` between the two models. While the liberal model expects attitudes toward climate action to be influenced substantially by individuals' beliefs regarding the existence and nature of climate change, these influences are absent or trivial in the conservative model. Instead, we observe that concern about climate change (`CC Worry`) is the only large cross-interaction toward `CC Action` in the conservative model.

Comparing now with the complete model (calibrated to the full dataset), we find that interactions absent from the complete model are generally also absent from both smaller models. The exceptions are $\{$`CC Others Worry`, `Weather Worry`$\} \longrightarrow$ `CC Action`, which are present in the conservative model, albeit with small effect sizes.

The converse does not hold; in several cases edges are absent from the smaller models, yet included in the complete model (e.g., `CC Human` $\longrightarrow$ `CC Action` in the conservative model, `CC Worry` $\longrightarrow$ `Weather Worry` in the liberal model). While it is tempting to interpret these as differences in the ideology-specific belief systems, this inference is not necessarily valid. Given the smaller datasets used to calibrate these models—and the slow-moving dynamics of the measured variables—these edges may be excluded on the basis that we do not observe their effects. This is less likely for effects which are more substantial in the complete model, and thus supported better by the dataset.

Chapter 8

Discussion

Key points to hit:

3. Interpret the results in relation to the research questions. What do they mean in the context of the study? How do they support/conflict with previous research?
4. Discuss limitations
5. Implications of findings. How can the results be applied practically? What questions/ directions do the findings suggest?
6. Conclude with summary of the main points and reiteration of our contributions

In this chapter we interpret the results of the previous two chapters with respect to the research questions posed back in Chapter 2 and restated below, as well as the broader context of belief system dynamics. We then describe limitations of the present study, and conclude the chapter with a discussion on the implications of our findings for future work on belief system dynamics and belief-level interventions.

RQ1: To what extent are causal relations *symmetric* or *asymmetric*, in models of climate change belief systems inferred from the climate beliefs dataset?

RQ2: How do asymmetric and symmetric beliefs systems differ with regards to intervention strategy and effectiveness, in models inferred from the climate beliefs dataset?

RQ3: How do intervention outcome and effectiveness vary between individuals with different initial conditions in asymmetric belief systems inferred from the climate beliefs dataset?

RQ4: How do asymmetric belief systems inferred from the climate beliefs dataset vary between conservative and liberal individuals?

On the question of existence, Chapter 6.1 presented evidence of asymmetric directional relations between several pairs of beliefs or attitudes, while also finding that not *all* pairs necessarily exhibit this asymmetry. Two attitudes in particular—political ideology/alignment and climate-related concerns—displayed significant preponderance of influence over several other beliefs and attitudes.

Following this, Chapter 6.2 demonstrated that relational symmetry assumptions meaningfully impact intervention dynamics and outcomes. Interventions on political ideology/alignment in the asymmetric model were found to be almost universally more effective than those in the corresponding symmetric model. For inbound interventions targeting attitudes toward climate action, the results showed differences in relative effectiveness for different points-of-intervention between the two models. In one case, the differences were sufficiently large to change intervention effectiveness rankings, impacting intervention strategy.

We then changed tack to investigate heterogeneity in intervention effects and belief systems. Chapter 7.1 found that intervention effectiveness varies predictably with respect to individuals' pre-intervention belief states. Most high-effect interventions targeting attitudes toward climate action required low initial values for both the point-of-intervention and target. Surprisingly, pre-intervention climate-related concern was required to be low for *all* of the personas identified as characteristic of effective interventions, even when this was neither the point-of-intervention, nor the target.

Finally, Chapter 7.2 showed substantive differences between asymmetric belief system models calibrated separately to conservative and liberal subpopulations. Each model featured high-magnitude interactions not present in the other, yet while the liberal model was considerably sparser than the conservative model, most interactions in the former were also present in the latter. In comparison with the asymmetric model calibrated on the complete dataset, used in the prior experiments, the ideological models were both sparser, yet together had high overlap in connectivity with the complete model.

Interpreting asymmetric relations:
- In terms of the mathematical model
- Compare with other notions of 'influence' or 'centrality'

Political ideology/alignment and climate-related concern as asymmetric instigators:
- Interpret
- Compare with other results

Explaining differences in intervention results:
- In terms of influence/influentiability
  - Distinction which isn't possible in the symmetric model
  - Also comes back in the heterogeneity results
- Model misspecification

Measurement timescales:

Answering the research questions:
- Ideally 1–2 sentences for each

Interpreting 'with respect to the broader context of belief system dynamics':
- Likely one paragraph

8.1 RQ1: Asymmetric influence of political ideology & climate concern

Our findings in Chapter 6.1 demonstrate, by example, a positive response to the question of asymmetric relation existence. Two variables in particular—political ideology and climate-related concerns—exhibited significant asymmetric influence over the behaviour of several other variables. Additionally, we observed two forms of asymmetry: where two beliefs or attitudes reinforce one another with different strengths, and where one belief or attitude influences another with no apparent direct reinforcing feedback.

However, not all pairs of beliefs or attitudes display asymmetry. Aside from interactions with these two variables, we found that most interactions were either symmetric or near-symmetric, or otherwise have inconclusive support for asymmetry due to sampling error. Moreover, the presence or absence of asymmetry for different variables was not cleanly explained in terms of differences in average influence (interaction strength) or sampling error.

Hence while asymmetric relations between beliefs and attitudes appear to be less prevalent than symmetric ones (at least in this context), they exist nonetheless.

As per the underlying mathematical model defined in Chapter 3, we interpret belief relations temporally, as the influence of one belief or attitude on the future state of another. An asymmetric relation is one in which one belief or attitude influences (i.e., constrains) the other more than in the opposite direction. Our findings therefore suggest that in several cases political ideology/alignment and climate-related concerns drive the behaviour of other beliefs and attitudes, while being relatively insensitive, themselves, to the states of those beliefs and attitudes.

For instance, political ideology and/or alignment exerts asymmetric influence on general attitudes toward action on climate change. This is to say that a conservative and/or Republican-aligned individual is more likely to display lower *future* support for climate action than a liberal and/or Democrat-aligned individual. Yet conversely, if an individual's support for climate action changes, this asymmetry implies a smaller corresponding shift in their political views.

TODO:

- Comparison with other work:
  - Politics driving support for climate action, belief in CC, concerns about CC, while not being substantially influenced by these
  - Climate concern driving beliefs about climate impacts and causes, while not being influenced so much by them (e.g., psychological distance)
  - Other discussions on variables with varying 'influence', e.g., centrality.

8.2 RQ2: ...

While we have demonstrated the existence of asymmetic relations between beliefs and attitudes, existence does not, on its own, imply that the observed asymmetric relations are impactful on endogenous dynamics. Our findings in Chapter 6.2, however, show that assumptions regarding asymmetry *are* consequential when reasoning about interventions.

We showed, by way of simulation, that asymmetric models calibrated to the climate beliefs dataset exhibit post-intervention behaviour which differs from the corresponding symmetric model. Interventions on political views in the asymmetric model outperformed those in the symmetric model almost universally across target beliefs and attitudes. This was further reflected in our experiments on inbound interventions targeting attitudes toward climate action. Under the symmetric model assumption, intervening on beliefs regarding the impacts of climate change is expected to be more effective than intervening on political views, while under the *asymmetric* model asummption the situation is reversed.

One of the symmetric model's key strengths is its small(er) parameter count. With fewer degrees of freedom than the asymmetric model, it can achieve more accurate parameter estimates when calibrated to the same dataset (i.e., smaller confidence intervals in

Figure 5.2). However, this comes at the cost of model misspecification when a subset of the true relations are asymmetric. In the symmetric model, bidirectional interaction strengths are generally inferred as being between the corresponding directional interaction strengths in the asymmetric model.

This has little impact on beliefs and attitudes with largely symmetric relations, but can dramatically affect the level of influence for those with asymmetric relations, and hence affect intervention dynamics considerably. For instance, consider again the attitude associated with political ideology/alignment. In the asymmetric model this displays similar outbound interactions as the (largely symmetrically-interacting) belief regarding the impacts of climate change. On the other hand, in the symmetric model most interactions with political ideology/align are roughly half the magnitude of those for the latter belief. The result is a diminishing of the influence of this attitude, due to model misspecification, causing the latter belief to appear more influential, when, in-fact, it is almost identical. Regularisation can exacerbate this issue. In the asymmetric model, political ideology/alignment has outbound interactions with all (seven) other beliefs and attitudes. In the symmetric model, only four remain.

For the measurement timescale used in the experiments (roughly 2.5 years) we found that interventions act mostly through direct relations, as evidenced by rough correspondence between intervention effects and direct interaction strength. While some indirect influence was observed (e.g., for intervention targets with no direct connections to the point-of-intervention) this was small compared to direct effects, though increases for longer measurement timescales. This finding is potentially impactful for the difference between symmetric and asymmetic intervention dynamics. If two beliefs or attitudes are related asymmetrically such that only one directional interaction is nonzero, then a symmetric model is likely to both *overestimate* the effect of an intervention on the more influential variable, and *underestimate* the time until the effect is realised.

In summary, and responding to **RQ2**, we find that assumptions regarding the symmetry or asymmetry of relations between beliefs and attitudes can impact expectations regarding both the absolute and relative effect of intervention, as well as the time taken for the effects of an intervention to be realised. Differences between the models' dynamics arise both as a direct result of asymmetric relations existing (e.g., as in the example from the previous paragraph), and due to differences in influence (outbound interactions) and influen*tiability* (inbound interactions) which are characteristic of beliefs and attitudes in asymmetric relations. The latter can, from the other side, be considered a difference resulting from model misspecification — assuming that a belief system comprises only symmetric relations, when asymmetric relations are present.

8.3 RQ3: ...

In Chapter 2 we hypothesised that in order for interventions to be effective, it is necessary for the initial states of both the point-of-intervention and target beliefs or attitudes to be different from the desired post-intervention states.

The results in Chapter 7.1 broadly support this hypothesis for interventions targeting attitudes toward climate action, showing that the most effective interventions are observed in cases where the point-of-intervention is initially 'low', and initial attitudes toward climate action are negative. We found no requirement that the initial attitudes be close to the low extreme. This is consistent with our earlier discussion in Chapter 3.5, in which we illustrated how the impact of an intervention on the *point-of-intervention* behaviour diminishes as the magnitude of the initial influence on that belief or attitude grows (i.e., for strong opposition or support).

Unexpectedly, for all points-of-intervention, the majority of individuals for whom we estimate a high intervention effectiveness also had low pre-existing concerns about climate change. This was true even when climate-related concern was neither the point-of-intervention nor the target. This result is likely related to the variable's high degree of influence in the asymmetric model, in combination with a high potential to *be influenced*, compared with other variables.

Overall, we find that intervention effectiveness does vary in predictable ways with respect to individuals' pre-intervention belief system state. Effectiveness depends not only on the initial states of the target of point-of-intervention, but also on other auxiliary beliefs and attitudes through which interventions can propagate indirectly.

8.4 RQ4: ...

Plan:

1. **A:** What have we learned?

   Findings, per-research-question:

   RQ3: *Individual impacts*

   - **A:** In line with our hypothesis, for interventions (targeting `CC Action`) to be highly effective, it is generally necessary that both the target and point-of-intervention states be low (such that the intervention changes the point-of-intervention, and there is space for the target to shift)

- **B:** Unexpectedly, almost all highly effective interventions were also predicated on a low initial level of climate-related concern.
- **T:** We attribute this to climate-related concern's high degree of influence in the model, in combination with a high potential to *be influenced*, compared with other variables.
- **T:** Intervention effectiveness is not only dependent on the initial states of the target and point-of-intervention, but also by other auxiliary beliefs and attitudes through which interventions can propagate indirectly.
- *Comparison with other work:* Perhaps we have sources discussing the impact of climate concerns on policy specifically; ideally can find sources talking about interventions.

RQ4: *Individual belief systems*

- **A:** Conservative and liberal belief system models differ in sparsity (edge existence) and strength of specific interactions.
- **A:** The liberal model is considerably sparser than either the conservative model or the model calibrated on the complete dataset.
- **A:** In several cases, high-effect interactions are present in only one of the two models.
- **B:** The ideological models do, however, still display substantial overlap in the inferred edges.
- **T:** Belief systems may differ significantly between individuals or subpopulations at the level of individual relations, but nonetheless appear to exhibit some shared structures.
- *Can we find evidence, explanations for some of the big differences between the models?*
- *Compare to other research*

2. **B:** Limitations

How we think about interventions:

- **A:** We have assumed that interventions can act directly, and with equal effect on different beliefs and attitudes. This is to say, we are not concerned here with the nature of an intervention itself (the interface between the intervention and the belief system). Rather, we operate under the assumption that we *can* intervene, and study the resulting endogenous dynamics.
- **B:** In reality, some beliefs or attitudes may be easier or harder to intervene on than others.

- **T:** Taking the intervention process into account may result in different expected effects of intervention. For instance, we found that political ideology is often influential, but is difficult to influence, due to a scarcity of incoming interactions. Supposing, then, that we can only intervene indirectly on politics, the expected intervention outcomes may change.

Representational limitations:

- Polar vs. binary states:
  - We use polar $\{-1, +1\}$ states. This means that whatever the state of a spin, it exerts nonzero influence on spins it relates to. This makes more sense for some variables than others. For instance, political alignment and measures of policy support have clear positive and negative states (up to re-labelling). Beliefs such as 'Climate change is real' are murky, and whether they fit this assumption depends on how the survey questions are framed. In particular, we should distinguish between holding the opposite belief, and holding *no belief*. In the latter case (e.g., in questions such as 'Do you believe that …'), there is an argument to be made that the absence of belief should impose no influence on other spins.
  - Relates to discussion on zero (van der Maas et al., 2026)
- Vaccination example: Pairwise relations can't capture mediated relations between beliefs. i.e., relations which depend on other beliefs.
  - Can tie this into the individual belief systems limitation

Individual belief systems:

- **A:** We see differences between the ideological models which suggest that while individuals may share some structural components of their belief systems, the existence and strength of interactions can vary on an individual basis.
- **A:** This has also been discussed at length in (Brandt & Morgan, 2022).
- **B:** Estimating individual-level belief systems remains an open problem. Limited data, and the fact that many transitions will not be observed. Some progress has been made for bidirectional belief systems, e.g., Brandt (2022).
- **A:** In our experiments we have made the simplifying assumption that belief systems are shared.
- **B:** As seen in the comparison of symmetric and asymmetric model dynamics, the structure and relative interaction strengths in a belief system can change endogenous dynamics.
- **A:** The complete model can be seen as a 'mixing' of the two ideological models.

- **T:** We would expect to see differences in intervention dynamics between individuals to a greater extent than observed in RQ3. Yet expect that the observations from the complete model reflect an 'average case'.

Synchronous updates:
- **A:** We use synchronous updates to simulate model dynamics.
- **A:** This is due to both the nature of the dataset (long intervals between measurements, such that multiple beliefs can change state) and computational reasons (synchronous updates simplify the sampling process (Nguyen et al., 2017))
- **B:** If beliefs update asynchronously (one at a time), this could affect the model dynamics

Confounding factors:
- Model calibrated to only two waves. Cannot separate endogenous from exogenous factors. Individual-level exogenous factors likely to be 'washed out'. Shared factors (e.g. election, weather events) could cause correlated changes between individuals.

Transferring natural endogenous dynamics to intervention:
- Related to confounding factors, we (assume that we) calibrate the models to data which is taken from a 'normal' environment.
- Possible that dynamics are different when we intervene. Raises 'temperature'. Causes to think about beliefs/attitudes that would otherwise remain dormant—conflicting but unnoticed.

Data variables:
- We use what is available, rather than what is ideal

Sensitivity to unmeasured factors or incorrect structure:
- What happens when we cannot include (because we don't measure) an influential belief, i.e., a fork
- What about colliders, or paths?
- What happens when we falsely assume that there is/isn't a connection between two beliefs?

3. **T:** Open questions and future work
   - Individual belief systems
   - Representational capacity
   - Purposeful intervention study
   - Index variables could be derived intentionally, i.e., theoretically-motivated.

Implications of findings. How can the results be applied practically? What questions/ directions do the findings suggest?

- **T:** The symmetric assumption may often be valid, or at least a reasonable approximation (e.g., when asymmetry exists, but the difference in effects is small), yet asymmetric relations are likely to exist, and will not be captured by such an approach.

Conclude with summary of the main points and reiteration of our contributions

1. **A:** Contributions *Directed, causal belief system model*

   Longitudinal data, captures dynamics as opposed to observed state

   RQ-focused contributions

Chapter 9

Climate beliefs dataset

TODO:

- Create table describing climate beliefs dataset variables
- Discuss how we create the index variables

In this chapter we outline the context and construction of the dataset used for model calibration in Chapter 4. We will first describe the broader context and complexities of the dataset underlying this study, before outlining our data validation and cleaning methods in Chapter 9.2. Finally, in Chapter 9.3 we detail the construction of the targeted dataset used for model calibration (Chapter 4).

While many empirical studies on belief systems rely on cross-sectional data (Lee et al., 2025; Powell et al., 2023; van Noord et al., 2025), longitudinal data is necessary to capture changes in individuals' internal cognitive states over time. In fact, repeated per-individual observations are essential to our adopted method for model calibration. We are fortunate, in this study, to have had access to a longitudinal dataset on climate beliefs and attitudes in USA as our primary data resource.

The longitudinal study (**CITE**) comprises six waves of responses from individuals residing in the United States, collected between April 2020 and March 2023. The assessed dimensions include general demographic information (e.g., age, gender, education, and financial status), beliefs, attitudes, and experiences relating to concurrently-salient topics such as COVID-19, climate change, or the 2020 US presidential election, and support for hypothetical policies. In addition to a wide-form table of survey response data, the dataset includes a codebook which specifies, for each survey item, the question text, occurrence in survey waves, and conditional display logic where applicable. At present, the codebook is limited to Waves 1—5, i.e., excluding the sixth (final) wave.

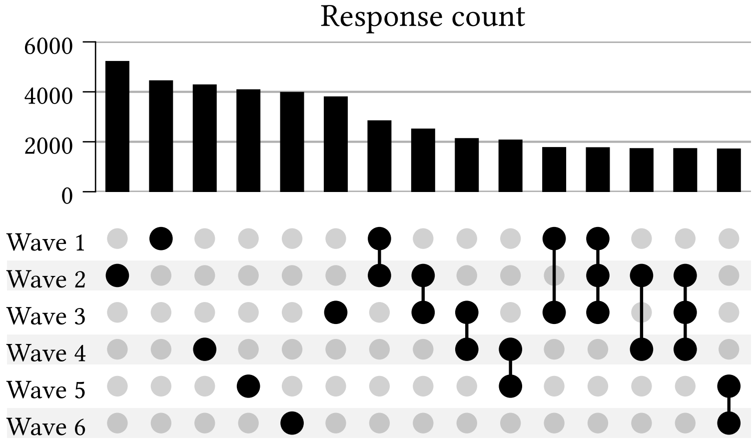


Figure 9.1: Number of repeat participants for different wave combinations. Combinations with total responses below 1500 have been removed.

However, the dataset is not without complexities. Survey participation varies, with each participant responding to a (possibly non-contiguous) subset of waves. Figure 9.1 shows the number of participants who have responded to each combination of survey waves, with a minimum count of 1500. While each individual wave has a relatively high response rate (roughly 4000–5000), this rapidly drops off when other waves are considered. For instance, only ~2500 individuals responded to Waves 1 and 2, and only ~1900 of these individuals also responded to Wave 3.

Survey content also varies between waves and participants, both with regards to *which* questions are included, and *how* they are presented. Certain questions are displayed only in particular waves, or only to either repeating or new participants. Some questions are shown conditionally based on an individual's responses to prior questions (within the same wave). Others vary according to survey treatment conditions, such that individuals in different treatment groups are presented different variants of a question. This survey logic is not always executed correctly; in some cases participants are shown survey questions incorrectly (e.g., 'new' participants shown questions intended only for 'repeating' participants). We discuss this issue further in Chapter 9.1.

Question format, text, and response schemas also occasionally change between survey waves. Changes in response schema are less common, however, and typically affect items with categorical responses. With a view to constructing the calibration dataset for our experiments, we note that most response schemas are not binary and therefore require binarisation (see Chapter 4.2).

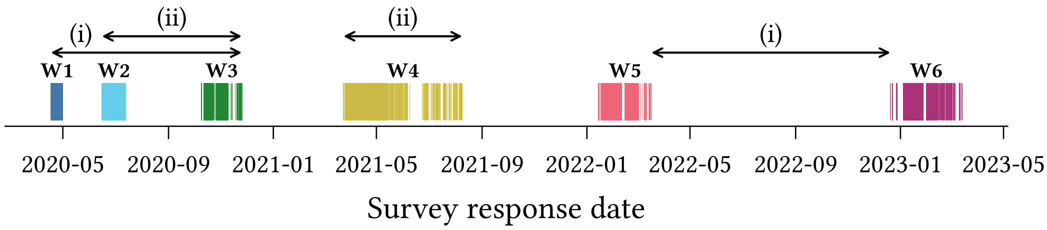


Figure 9.2: **TODO**

Casting our attention to the survey timing, we examine the survey response dates for each wave (Figure 9.2) and the distribution of interval durations between consecutive-wave responses across individuals (Figure 9.3), i.e., the 'inter-response time'.

In Figure 9.2 we observe that the survey waves occur with irregular spacing and duration. Some consecutive pairs of waves are much closer than others. For instance, notice that:

(i) The duration from the start of Wave 1 until the end of Wave 3 is shorter than the duration *between* Waves 5 and 6, and

(ii) The duration spanned by Wave 4 alone exceeds that of the interval spanned by both Waves 1 and 2.

This poses a potential problem for model calibration, since the non-equilibrium belief model (defined in Chapter 3) operates on the assumption that samples are equispaced.

We see the irregular spacing reflected also in the inter-response time distributions. For each pair of consecutive waves, we observe substantial variation in the time between responses for different participants.

Finally, we note that the time interval spanned by the longitudinal dataset includes several notable events which could reasonably be expected to influence — and confound — the dynamics of beliefs and attitudes in myriad contexts. These include the COVID-19 pandemic, which arrived in the US only three months prior to the first survey wave

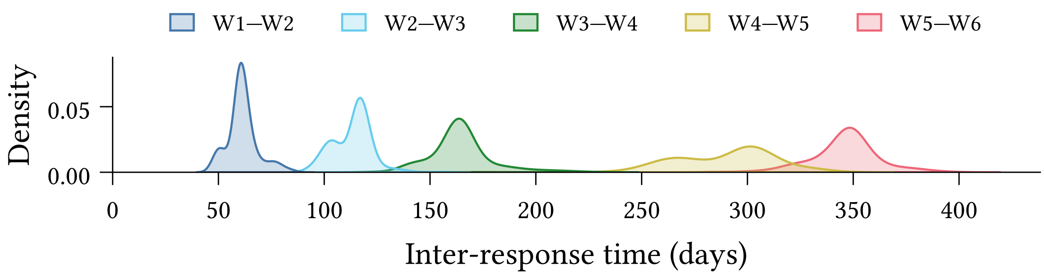


Figure 9.3: Between-response time distribution

(Holshue et al., 2020), the 2020 US Presidential Election which occurred during Wave 3, the January 6 United States Capitol Attack, which occurred between Waves 3 and 4, ...

9.1 Data Validation

The complexity of the climate attitudes survey, as outlined above, necessitates a rigorous approach to data validation, both to identify errors in the expected schema and to ensure that the data matches our expectations.

We have implemented a general validation pipeline comprising three stages:

1. **Type-level validation:** The set of columns is exactly as expected, and all columns have the correct data type.

2. **Response-value validation:** All *non-null* responses are valid according to the schema specified in the codebook.

3. **Null-value validation:** Responses are null if, and only if, we expect them to be null.

9.1.1 Type-level validation

Type-level validation ensures that the *observed* data schema matches the *prescribed* schema. Since response data types vary between questions (see Table 9.1), the type-checking must be flexible and capable of handling complex data types.

For instance, categorical multiple-choice responses are represented using a `list[Enum]`, where `Enum` is a question-specific enum-type[13]. Type validation for a multiple-choice question thus requires checking: (i) that the response column comprises `list`'s, and (ii) that all list elements belong to the set of values defined by the `Enum`.

9.1.2 Response-value validation

Response-value validation then ensures that all non-null response values are valid according to the survey codebook. For most variables this is straightforward. Numeric and single-response ordinal variables typically have a defined range (e.g., $18 \leq \text{age} \leq 99$,

[13]An enum is a type defined by a finite set of allowable values. In our case the values are human-readable strings. For instance, the `dem_urban` survey question, which asks 'What kind of area do you live in?' has responses with data type described by the enum $\{\text{Urban}, \text{Suburban}, \text{Rural}\}$.

Response schema	Raw type	Coerced type
Text	`str`	`str`
Single response (ordinal)	`int`	`int`
Single response (categorical)	`int`	`Enum`
Multiple response	`str`	`list[Enum]`
Numeric	`float`	`float | int`

Table 9.1: **TODO**

or $1 \leq x \leq 5$ for a 5-point Likert scale variable). Text-entry responses are always considered valid.

Single-response categorical questions have `Enum` type, so type-level validation is sufficient to ensure that the responses do not contain any values outside the set defined by the `Enum`. However, the allowable values occasionally change between survey waves. Hence response-value validation is also required for categorical single-response and multiple-response variables, and in general must handle schema variation between waves.

9.1.3 Null-value validation

Finally, null-value validation ensures that responses are null if, and only if, they are expected to be null. For a question Q, the response of a participant P in wave W is permitted to be null, iff, at least one of the following four conditions is true:

N1. Q is not asked in wave W,

N2. In wave W, Q is only shown to *new* participants, but P is a *repeating* participant (or vice versa),

N3. In wave W, Q is only shown to a treatment group of which P is not a member, or

N4. In wave W, Q is conditionally shown to participants based on their response to a distinct question Q', and P's response to Q' does not satisfy the condition.

Formally we can write the condition for P's response R being null as:

$$\text{null}(R) \iff (\text{N1} \vee \text{N2} \vee \text{N3} \vee \text{N4}) \tag{9.1}$$

Null-value validation then consists in identifying the set of responses $\boldsymbol{R}$ such that for $R \in \boldsymbol{R}$, exactly one of null(R) and (N1 ∨ N2 ∨ N3 ∨ N4) is true. Validation succeeds if $\boldsymbol{R}$ is empty. Otherwise the members of $\boldsymbol{R}$ are counterexamples to the above conditions.

9.1.4 Implementation

We implement type-level and simple response-value validation using the Pandera Python library for DataFrame validation (Bantilan et al., 2022). We implement manual validation checks for more complex response-value cases, such as questions whose response schema varies between waves.

The null-value validation logic, i.e., the process of testing for contradictions Equation 9.1, is also manually implemented. Conditions N1 and N2 are derived automatically from the survey codebook. Conditions N3 and N4 must be manually specified, as the codebook currently does not have a standardised method for describing conditional display logic.

9.1.5 Validation results

At the time of writing, approximately 25% of the complete survey question set for Waves 1–5 has been validated, including all survey questions which were considered for the targeted calibration dataset (Chapter 9.3). Since Wave 6 is currently undocumented in the survey codebook we can reasonably validate neither the data schema, nor the presence of null-values. We have therefore not yet validated any of the data from Wave 6, and exclude this wave from the present study.

All type-level and response-value validation checks succeed, providing a strong guarantee that the data schema matches our expectations per the codebook. We do, however, encounter several problems during null-value validation.

In some cases, these were due to errors in the codebook itself. These errors are relatively straightforward to identify from the null-value validation results, since they often affect all individuals in a particular wave. For instance, if the codebook specified that a question is not presented in Wave 1, yet all responses are non-null, then this most likely indicates an error in the codebook. While some cases are more subtle, such as where a treatment class is misspecified, when arising due to a codebook error we still expect the contradictions to affect a well-defined subset of the population (in this case, the individuals in a particular treatment class).

In other cases we identify contradictions which are due to errors in the survey process, which caused certain survey questions to be displayed to some individuals in conditions which did not satisfy the specified survey logic. Some such cases were systematic, affecting all individuals until the survey process was updated; however, in other cases we have not been able to identify the source of the error.

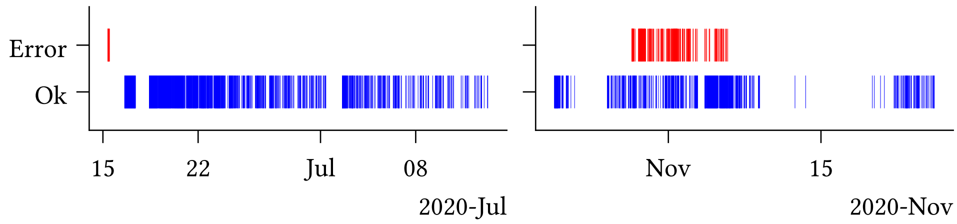


Figure 9.4: **TODO**

We exclude all responses which fail the null-value validation from further analysis in the data cleaning stage (Chapter 9.2.2).

The described validation process is primarily concerned with ensuring that the data schema — comprising types and values — aligns with our expectations as per the survey codebook. While comprehensive in this regard, the process does not account for *all* possible forms of errors or inconsistencies. One category which is currently unaccounted for, but is worth mentioning, comprises inconsistencies in the responses from a given individual over multiple waves. For instance, the dataset includes several cases in which individuals presented the question:

"Were you born in the United States?"

respond with 'Yes' in one wave, but 'No' in another. Validating data for inconsistencies such as this requires careful consideration on a question-by-question basis to assess response types for conflicts. We consider this category beyond the scope of this study, describing it here only to illustrate the bounds of the above validation process.

9.2 Normalisation, cleaning, and transformations

We apply pre-processing steps to make the dataset amenible to downstream analysis, which are divided into three categories:

- **Normalisation:** Ensuring the data schema (variable names, types, and values) conform to consistent standards.
- **Cleaning:** Filtering erroneous or otherwise-problematic data.
- **Transformation:** Structural dataset enrichments, including constructed data variables and improved organisation of information.

9.2.1 Normalisation

Variable names in the original dataset are subject to some inconsistency, featuring a mixture of `snake_case`, `CamelCase`, `dot.case`, and `kebab-case`. We therefore first convert the names of all variables specified in the codebook to a standard format, making all names lowercase, and replacing hyphens and periods with underscores.

The dataset also contains several columns which are not described in the codebook, often containing survey metadata, the functions of which are typically indicated using a variable name prefix. For instance, the `Group_` prefix indicates that a column describes treatment group membership. We do not normalise these variable names here, so as to maintain identiability for later transformations (Chapter 9.2.3).

Secondly, we coerce all variable data types to a standard set. In most cases this is straightforward. Dates and times represented as strings are coerced to timezone-free `datetime` types. Non-floating-point numeric values (which constitute most survey response types) are coerced to 64-bit integers.

Categorical variable responses are originally represented as integeres, which presents three issues. Firstly, it imposes an ordinal structure which does not, in general, make sense for categorical variables. For instance, consider the first four response options to a survey question asking participants about their recent experience with different forms of extreme weather:

1. Severe winter storm, 2. Hurricane, 3. Tornado, 4. Heat wave

One may devise any number of reasonable ordinal interpretations over this set (e.g., 'highest average risk of property damage', or 'annual emergency response cost'), yet such interpretations are context-dependent and not inherent to the categories themselves. Recording categorical responses using numeric types therefore risks spurious interpretations. Second, integer values provide no information about the categories to consumers of the dataset, requiring that they cross-reference manually with the codebook. Thirdly, in some cases the mapping from categories to integers for a given question varies between survey waves. We account for all three of these issues by coercing categorical variables to `Enum` types, which preserve information about the underlying categories, and do not assume an ordinal structure.

Multiple-selection survey items are more complex still. These are represented as strings comprising comma-separated integers, whose values typically refer to non-ordinal categories. We coerce these to `list[Enum]` types, which have all the benefits of the categorical enums discussed above, and enable simpler programmatic analysis (e.g., testing for set membership, determining differences in response sets between waves).

Finally, we perform a series of value updates to ensure consistency between responses to different questions. This consists in replacing all empty string responses with `null` values[14], and re-mapping integer columns (both numeric and ordinal) such that the minimum value is zero.

9.2.2 Cleaning

Following schema normalisation, we then perform a cleaning stage, which primarily includes filtering out invalid or problematic data. We only filter out entire survey responses, rather than answers to specific survey items. Survey responses may be removed for any one of three reasons.

Firstly, we remove responses where the `participant_id` column is `null`, such that the individuals cannot be traced across survey waves. Secondly, we remove any responses from participants who fail the survey validity check in *any* of their participation waves. Thirdly, we remove any responses which fail either of the response-value or null-value validation checks described in the previous section (Chapter 9.1). We recognise that these issues are not necessarily problematic in every research context (e.g., cross-sectional studies are unconcerned with tracing an individual's responses across waves), and thus have made each condition optional in the dataset construction code.

9.2.3 Transformation

Finally, we apply several transformation steps with the goal of simplifying manipulation and analysis of the existing dataset, or enriching it with additional information.

The original dataset includes two survey items querying participants' alignment with the two largest US political parties (i.e., the Republicans and the Democrats). The first item asks whether the participant whether they consider themselves as *Republican*, *Democrat*, or *Independent*. Those who respond *Independent* are then presented a follow-up question asking whether they *lean* toward either (or neither) party. We replace these two survey items with a single variable combining the responses, with possible values:

$$\text{Democrat} \prec \text{Leaning Democrat} \prec \text{Independent} \prec \text{Leaning Republican} \prec \text{Republican}$$

[14]Note that this particular value mapping must happen *after* null-value validation (Chapter 9.1).

While this supplements the first item with additional information, we note two potential limitations of this transformation. Firstly, some participants who respond *Republican* or *Democrat* to the first question may more accurately describe themselves as only leaning in a given direction, if they had been aware of the (conditional) second question. Therefore the 'Leaning' responses are perhaps better interpreted as 'Independent with Republican/Democrat tendencies'.

Secondly, our implicit assumption that *Independent* is a midpoint between *Republican* and *Democrat* would misrepresent individuals who consider themselves 'right-of-Republican' or 'left-of-Democrat'. However, the available data does not allow us to distinguish such cases. Furthermore, this issue is also present in the original first question.

Our second transformation concerns survey questions with different variants, which are conditionally displayed based on treatment group membership. The original dataset includes, for each treatment group associated with a given question, a separate response column, as well as a binary indicator column describing group membership. Since group membership is always mutually exclusive in the dataset, we coalesce these various columns, replacing them with singular response and group membership columns. The group membership column contains integer indexes as opposed to the original binary indicators. In some cases, different treatment groups have an associated numerical value. For instance, the `ccSolveX` items query support policies compensating communities affected by climate change, with the treatment group specifying the cost (X) of such a policy to the participant. In these cases we supplement the dataset with an additional column containing the associated numerical value, simplifying downstream analyses.

Finally, we improve the dataset organisation by constructing a database comprising tables for survey item and question metadata and responses, as well as participants themselves (Figure 9.5). The requirement for such as database arose during the construction of the climate beliefs dataset used for model calibration (Chapter 9.3). The original data, comprising a codebook (a spreadsheet) and an RData file containing rows of participant responses, is not immediately amenible to our task, which requires reasoning about both survey logic and participation in tandem. The constructed database primarily serves to link survey metadata (extracted from the codebook) with survey response data.

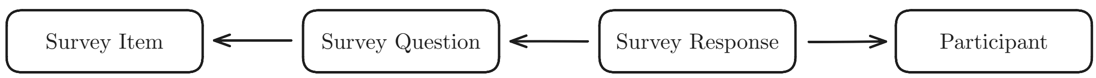


Figure 9.5: The survey database comprises tables for survey items (underlying concepts assessed in the survey), survey questions (the particular mode in which an item is assessed in a given wave or treatment condition), survey responses, and participants.

Recognising the hierarchical structure of the longitudinal dataset, in which a single survey item may vary in question text or response schema between waves or treatment conditions, we separate the concepts of *survey items* (the underlying concept being assessed) and *survey questions* (the particular mode of assessment). Survey questions are associated with particular waves, participant types (i.e., new or repeating), and treatment conditions, but may be related under a common survey item. *Survey responses* are associated with a particular question as well as a *survey participant*. We include a separate table for participants which records wave participation as well as the initial wave in which a participant has responded.

The database is stored on disk as a collection of parquet files (*Parquet*, n.d.), which preserve complex data type information and are supported in both R (e.g., using Arrow) and Python (e.g., using Pandas or Polars).

9.3 Climate beliefs dataset

In light of the complexities and breadth of content of the **placeholder dataset**, we construct a smaller, targeted dataset of beliefs and attitudes relating to climate change, which we expect – on theoretical grounds – to exhibit interdependent behaviour. We will refer to this targeted dataset as the **climate beliefs dataset**. The primary motivator for constructing this dataset is the calibration of the non-equilibrium belief system model (Chapter 4.3).

We aim for 7–10 variables in the final set, so as to keep both the number of model parameters and parameter uncertainty sufficiently small. We first filter the survey items to consider only those which assess cognitive states (e.g., beliefs, attitudes, opinions, stances). We further restrict our attention to items which either relate directly to climate change, or which are expected to influence climate-related cognitive states (e.g., political alignment).

Within this restricted set of survey items our intention is to select a collection of variables which: (i) exhibit related behaviour, (ii) maximise the number of observations, and (iii) satisfy the requirements of the parameter estimation method used for model calibration. We will discuss the latter two requirements now, and return to the first shortly. We require at least two waves of responses from each participant, as the model is defined by a time-dependent Markov process, and thus parameter estimation involves fitting a conditional probability distribution where each belief state depends on the previous one. Moreover, each participant's responses should be equispaced, with spacing being consistent across individuals, and the dataset must contain no missing values. In addition to these requirements, we prioritise including several variables of interest:

1. **CC Worry**: Concern about current and future climate change
2. **CC Others Worry**: Belief regarding *others'* level of worry about climate change
3. **CC Human**: Belief that climate change is caused by human activities

The first is generally considered influential in shifting related attitudes, while the second is expected to be influential on the first, as a perceived descriptive norm. Finally, while most inividuals in the USA believe in the *existence* of climate change, there is relatively greater variation in beliefs about it's *causes*. Hence the third item is an interesting target for intervention studies. (**TODO: Add citations**)

Given the above requirements and the objective to maximise the number of observations, we identify Waves 3 & 4 as suitable candidates. These waves include 1693 repeat observations (excluding survey errors as described in the previous subsection). After removing items with no substantial correlations, and small groups of items which exhibit only internal correlations, we identify 17 relevant survey items, comprising both beliefs (epistemic positions, Table 9.2) and attitudes (qualitative evaluations, Table 9.3).

To reduce the set of candidate variables to the target range of 7–10, we proceed to identify groups of similar or redundant variables which may be removed or combined into interpretable index variables. We use two primary measures to investigate similarity: pairwise partial correlation, and vector autoregression (Epskamp, van Borkulo, et al., 2018; Epskamp, Waldorp, et al., 2018).

Partial correlation measures the correlation between a pair of variables X and Y after conditioning on all remaining variables Z:

$$\rho(X, Y) = \mathrm{Corr}(X, Y \mid Z) \tag{9.2}$$

In particular, $|\rho(X, Y)|$ is non-zero if X and Y exhibit *linear* correlation which is not shared with the remaining variables in Z. For instance, consider the fork structure $X \leftarrow A \rightarrow Y$ where X and Y are probabilistic linear functions of A. If the value of A is unknown, then X and Y are dependent. However, once A is known, the dependency structure is broken. Thus $\mathrm{Corr}(X, Y)$ is nonzero, while $\rho(X, Y) = 0$.

Figure 9.6 displays the pairwise partial correlation between each pair of variables in the filtered set. Note that we have coded all variables such that the partial correlation matrix entries are predominantly positive. Cells with magnitude below 0.05 have been masked for visual clarity. Rows and columns are ordered using hierarchical clustering.

Item	Question text	Response schema
CC Real	Do you think that climate change is happening?	• Yes • No • Don't know
CC Human (*)	Do you think rising temperatures are a result of human activities, natural causes, or both?	• Not happening • Natural causes • Human activities • Both
CC Impact (world)	How much do you think climate change is currently harming the world in general?	• Not at all • Only a little • A moderate amount • A great deal
CC Impact (wealthy)	How much do you think climate change is currently harming wealthy communities within the United States?	• Not at all • Only a little • A moderate amount • A great deal
CC Impact (poor)	How much do you think climate change is currently harming poor communities within the United States?	• Not at all • Only a little • A moderate amount • A great deal
CC Impact (comm)	How much do you think climate change is currently harming your local community?	• Not at all • Only a little • A moderate amount • A great deal
CC Others Worry	How worried do you think most Americans are about global warming/climate change these days?	• Not at all worried • Not very worried • Somewhat worried • Very worried

Table 9.2: * Re-mapped to: includes/does not include human activities

Partial correlations are limited, however, in that they capture only instantaneous (or 'contemporaneous') relations between variables, while the non-equilibrium belief system model is concerned with temporal relationships. Therefore we also examine the temporal and contemporaneous networks obtained using vector autoregression (VAR) with a $\delta t = 1$ time lag (Brandt & Williams, 2007). These networks are derived from the solution to the following regression problem, described in Epskamp, Waldorp, et al. (2018):

$$\begin{aligned} \boldsymbol{y}^t &= \boldsymbol{B}\boldsymbol{y}^{t-1} + \varepsilon^t \\ \varepsilon^t &\sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Theta}) \end{aligned} \tag{9.3}$$

Where $\boldsymbol{y}^t$ is the vector of observed variable values for each individual at time t. The *temporal network* matrix $\boldsymbol{B}$ comprises linear next-state predictors, while the *contemporaneous network* matrix $\boldsymbol{K} = \Theta^{-1}$ describes the pairwise conditional correlations

Item	Question text	Response schema
CC Worry	How worried are you about current and future global warming/climate change?	• Not at all worried • Not very worried • Somewhat worried • Very worried
Weather Worry	How worried are you about an extreme weather event or natural disaster happening to you personally in the next year?	• Not at all • Only a little • A moderate amount • A great deal
CC Responsibility	It is important that individuals take action on issues of climate change.	5-point Likert agreement scale
CC Scientists	Scientists with appropriate expertise should guide how we respond to climate change	5-point Likert agreement scale
Policy: ICA (*)	Do you favour an international agreement committing the USA and other countries to reduce their carbon emissions?	• Yes and legally binding) • Yes but *not* legally binding) • No
Policy: Tax fuel	How much do you support/oppose a tax on the production/distribution of carbon-based fuelds?	5-point Likert support scale
Policy: Auto	How much do you support/oppose stronger carbon emissions standard for car manufacturers?	5-point Likert support scale
Policy: Env. Reg.	Which statement comes closer to your views?	• Stricter environmental regulations cost too many jobs and hurt the economy • Stricter environmental regulations are worth the cost
Political affiliation (**)	In politics today, do you consider yourself a:	• Republican • Leaning Republican • Independent • Leaning Democrat • Democrat
Political ideology	What is your political ideology?	• Very conservative • Conservative • Moderate • Liberal • Very liberal

Table 9.3: * Re-mapped to: Yes/No

** Constructed from separate items for identification and leaning

which remain after accounting for temporal effects. The entries of $\boldsymbol{K}$ are analogous to

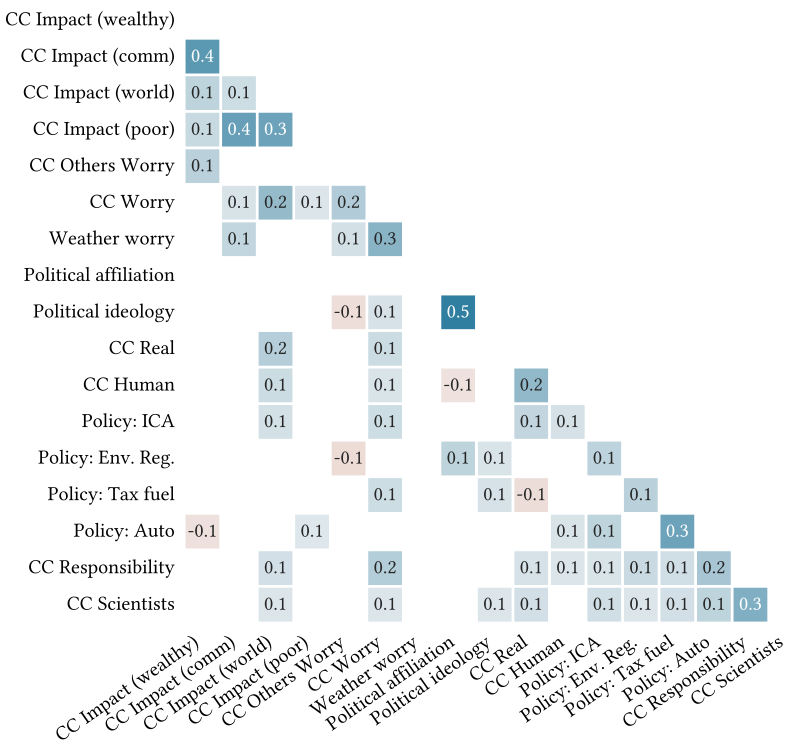


Figure 9.6: Pairwise partial correlation for the filtered candidate set (Table 9.2, Table 9.3). Cells with $|\rho(X,Y)| < 0.05$ are masked for visual clarity. Rows and columns are ordered using hierarchical clustering over the distance matrix: $d_\rho(X,Y) = 1 - |\rho(X,Y)|$.

partial correlations, constituting the conditional correlation between X and Y given the remaining variables.

Figure 9.7 shows the contemporaneous and temporal network matrices for the filtered set of candidate variables. The contemporaneous matrix is symmetric, thus the upper triangle is not shown.

We observe some apparent commonalities between the partial correlation and VAR networks. First, the *CC Impact* variables exhibit relatively strong internal relations. External relations with variables outside this set are typically weaker, with the exception of *CC Impact (world)*, which has several non-trivial partial correlations with other variables, including other climate-related beliefs. All variables in this group, except *CC Impact (wealthy)*, also have non-trivial partial correlations with *CC Worry* — one of our variables-of-interest. Second, the policy variables, as well as *CC Responsibility* and *CC Scientists*, also exhibit substantial connectivity in the partial correlation network. This is diminished in the contemporaneous VAR network, but visible in the clustering of the temporal network. The similarities among these variables in the rows and columns of the temporal matrix indicate that they fill similar roles as predictors, and are also

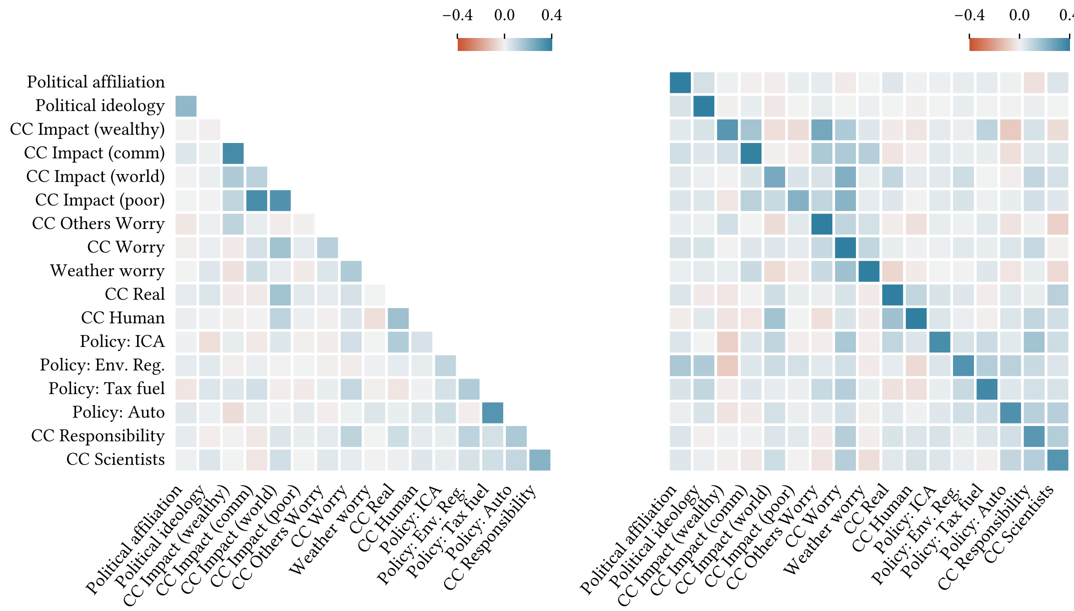


Figure 9.7: Contemporaneous (left) and temporal (right) network matrices for the filtered candidate set (Table 9.2 Table 9.3). Entries in row i of the temporal matrix are the regression predictors for the subsequent value of variable X_i. Rows and columns are ordered using hierarchical clustering on the distance matrix: $d_{\boldsymbol{K}}(X,Y) = 1 - |\boldsymbol{K}|$.

predicted similarly. Thirdly, *Political affiliation* and *Political ideology* are highly related in the partial correlation matrix, and also exhibit similar behaviour in the temporal network.

Based on these observations, we construct three index variables: (i) *CC Impact*, comprising the four climate impact beliefs, (ii) *Politics*, comprising the measures of political affiliation and ideology, and (iii) *CC Action*, comprising the four policy variables, as well as *CC Responsibility* and *CC Scientists*.

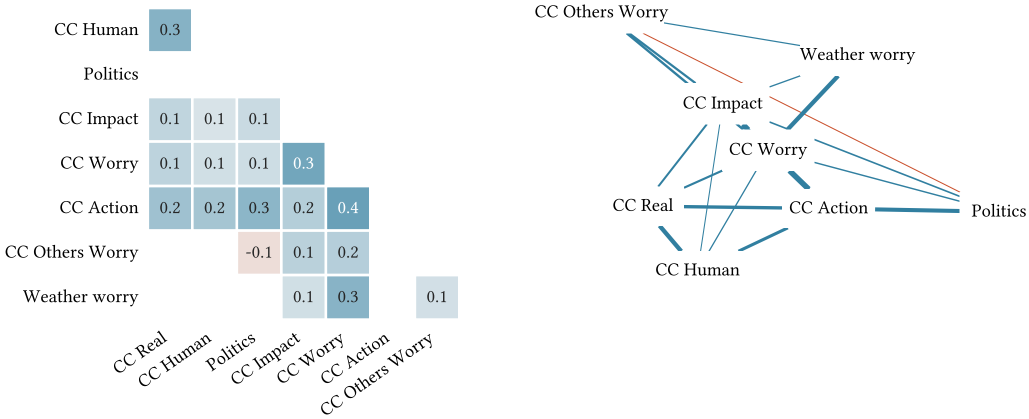


Figure 9.8: **TODO**

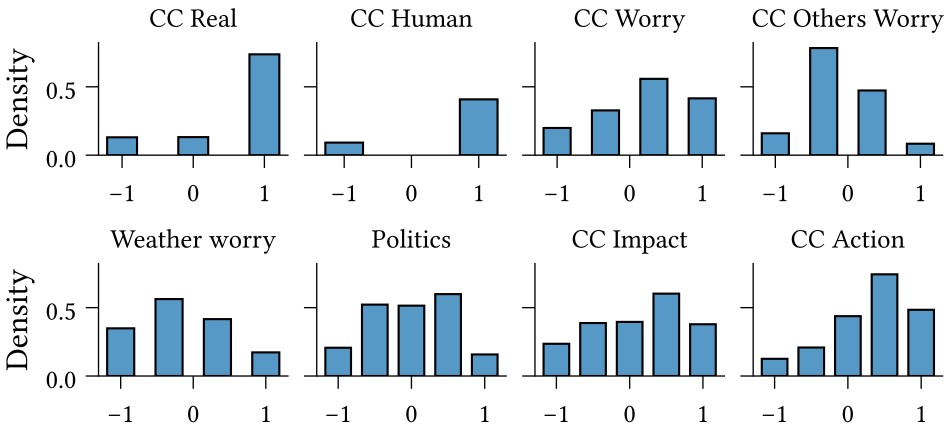


Figure 9.9: **TODO**

Chapter 10

Related work

10.1 Plan

1. Psychological/philosophical theories of beliefs and attitudes
   - Cognitive consistency:
     - Heider (balance theory) (Heider, 1946), Insko & Schopler (triadic consistency) (Insko & Schopler, 1967)
     - Festinger (cognitive dissonance) (Festinger, 1962)
   - Fishbein (Fishbein, 1966)
   - Quine (Quine, Willard van Orman & Ullian, Joseph Silbert, 1978)

2. Belief system modelling:

   - Schools of thought:
     - Triadic consistency (Rodriguez et al., 2016)
     - Parallel constraint satisfaction (Monroe & Read, 2008)
     - Bayesian networks (Cook & Lewandowsky, 2016; Powell et al., 2023)
     - Structural equation modelling?
     - Regularised partial correlation networks
     - Ising
       - CAN (Dalege et al., 2016)
       - Social influence: Network of belief (Dalege et al., 2025), hierarchical Ising model (van der Maas et al., 2020)

   - What theory/assumptions does each model draw on?

   - Difference between Latent variable and empirical network models (Dalege et al., 2016)

- Case studies — how have the models been applied?
  - Theoretical explanations for known phenomenon
  - Analysis of survey data

3. Interventions
   - Modelling, testing effects (Powell et al., 2023)
   - Theoretical (Dalege et al., 2025)
   - In-practice (The Workshop, 2024)
4. Inference
   - Between-individual limitations (Brandt & Morgan, 2022)

Chapter 11

Conclusions and future work

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Chapter 12

Ethics and Data Management

A new requirement for the thesis is that there must be a short section in which you reflect on the ethical aspects of your project. This requirement is related to one of the final objectives that a graduated student of the Master of Computational Science must meet: "The graduate of the program has insight into the social significance of Computational Science and the responsibilities of experts in this field within science and in society". You don't need to devote an entire chapter to this; a short section or paragraph is sufficient.

I acknowledge that the thesis adheres to the ethical code (https://student.uva.nl/en/topics/ethics-in-research) and research data management policies (https://rdm.uva.nl/en) of UvA and IvI.

The following table lists the data used in this thesis (including source codes). I confirm that the list is complete and the listed data are sufficient to reproduce the results of the thesis. If a prohibitive non-disclosure agreement is in effect at the time of submission "NDA" is written under "Availability" and "License" for the concerned data items.

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Appendix A

Derivations

A.1 Parameter estimation

A.1.1 Log-likelihood

Equation 4.5

A.1.2 Expected log-likelihood

Equation 4.6

A.1.3 Partial derivatives of expected log-likelihood

A.2 Dataset

A.2.1 Binarisation probability

Equation 4.1

Appendix B

Additional results

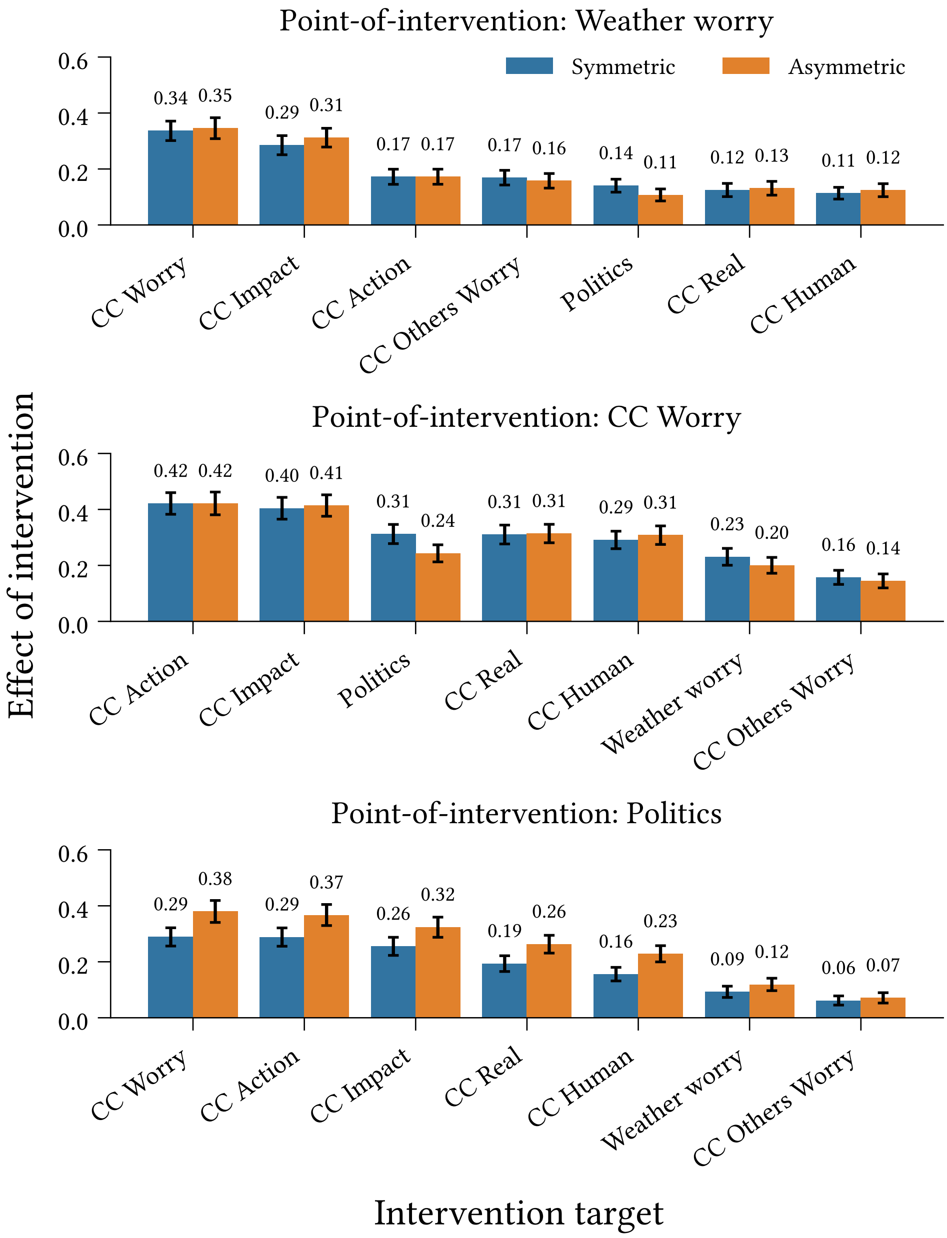

Figure B.1: Outbound effect of intervention (Definition 6.2.1) for interventions targeting `Weather Worry`, `CC Worry`, and `Politics`, in symmetric and asymmetric belief system models calibrated to the climate beliefs dataset. Measurements taken at $t = 10$ (approximately five years in the models' timescale). Intervention strength: $\delta_h = 2.5$. Confidence intervals display 1.96 standard deviations around the mean effect, measured across 500 repeated simulations.